



Tribhuvan University

Faculty of Humanities and Social Sciences

Real Estate House Price Prediction

A Project Report

Submitted To:

Department of Computer Application

Padmashree International College

In partial fulfillment of the requirements for the Bachelor in Computer Application

Submitted By:

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SUPERVISOR’S RECOMMENDATION

I hereby recommend that this project is prepared under my supervision by **Sanjay Kumar Khadgi** entitled “**Real Estate House Price Prediction**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application recommended for the final evaluation.

.....

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LETTER OF APPROVAL

This is to certify that this project is prepared by **Sanjay Kumar Khadgi** entitled “**Real Estate House Price Prediction**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application has been evaluated. In our opinion, it is suitable in the scope and quality as a project for the required degree.

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ABSTRACT

Nepal has experienced consistent growth in recent years, particularly in the real estate sector, which has been accelerated by urbanization and increasing demand for property. This research highlights the creation of a House Price Prediction Model that uses machine learning to meet this need by forecasting property values based on key factors such as location, area (sq.ft.), bhk, and bathrooms. A dataset from Kaggle was used, with a focus on real estate activities. Several machine learning techniques, including Multiple Linear Regression, Random Forest and K-Nearest Neighbor (KNN) were used to predict housing values, with approximate of 80% accuracy across the models. Multiple Linear Regression has the best accuracy of 83% making it the most effective model in this project.

This system has been implemented as a web application that allows users to enter relevant house information in order to receive price predictions. This system is especially useful for buyers and sellers to make informed decisions based on market trends. By analyzing the real estate market of Nepal, this system offers major insights on property values and pricing patterns.

Future work aims to further improve the model's accuracy by implementing additional features such as neighborhood quality and closeness to essential amenities, as well as experimenting with more advanced machine learning approaches. These enhancements could increase the model's precision and scalability, transforming it into a reliable tool for Nepal's real estate market.

Keywords: Linear Regression, K-Nearest Neighbors, Random Forest, House Price Prediction, Machine Learning.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
DB	Database
JSON	JavaScript Object Notation
JS	JavaScript
KNN	K-Nearest Neighbor
LASSO	Least Absolute Shrinkage and Selection Operator
MAE	Mean Absolute Error
MLR	Multiple Linear Regression
MSE	Mean Squared Error
MVC	Model View Controller
OOP	Object Oriented Programming
PLR	Penalized Linear Regression
PY	Python
REHPP	Real Estate House Price Prediction
REST	Representation State Transfer
RF	Random Forest
RMSE	Root Mean Squared Error
SVC	Support Vector Classification
SQL	Structured Query Language
SVR	Support Vector Regression
UI	User Interface
WLR	Weighted Linear Regression

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CHAPTER 1

INTRODUCTION

1.1 Introduction

House price prediction is a significant field within the domain of real estate and finance. It involves utilizing statistical model and machine learning techniques to forecast the prices of residential properties.

Accurately forecasting housing prices is of great significance to various stakeholders in the real estate sector, influencing decisions related to property transaction, investments and the formulation of policies. Traditional valuation methodologies often struggle with accurately capturing the complex relationships between many factors affecting house prices. This report explores the utilization of machine learning techniques, to predict housing prices. By analyzing a large dataset containing various housing attributes, this report aims to provide insights into the effectiveness of these machine learning methodologies in addressing the complexities of housing price prediction.

The utilization of machine learning techniques represents a significant opportunity to transform real estate forecasting, offering considerable potential for innovation and progress. Traditional valuation methods often struggle to adapt to the dynamic and complex nature of housing markets. Integrating machine learning models empowers analysts to navigate extensive datasets, uncover hidden patterns and identify nonlinear relationships among various housing attributes. Regression methodologies such as linear, k-nearest neighbor and random forest regression provide a systematic framework for understanding variable impacts on housing prices. Additionally, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are the preferred performance measure for regression tasks, because they effectively capture the intricate feature interactions. This project aims to leverage these machine learning methodologies to improve the accuracy and reliability of house price predictions, thereby transforming stakeholder engagement in real estate transactions.

1.2 Problem Statement

Prices of real estate properties are complexly linked with our economy. Despite this, we do not have accurate measures of housing prices based on the vast amount of data available. The general and standardized real estate characteristics are often listed separately from the asking price and general description. Because these characteristics are separately listed in a structured way, they can be easily compared across the whole range of potential houses.

Because every house also has its unique characteristics, such as a particular view or type of sink, house sellers can provide a summary of all the important features of the house in the description. All given real estate features can be considered by the potential buyers, but it is nearly impossible to provide an automated comparison of all variables due to the large diversity. This is also true in the other direction: house sellers have to estimate the value based on its features in comparison to the current market price of similar houses. The diversity of features makes it challenging to estimate an adequate market price.

1.3 Objectives

The objective of this system is to provide accurate estimates of property values, which are valuable for various stakeholders including homebuyers, sellers, investors and financial institutions.

- To develop a precise house price prediction model using machine learning techniques, integrating key attributes that affects the housing values.
- To analyze various parameters like area, locations, BHK, etc., to predict house prices accordingly.
- To compare the performance of different machine learning models for house price prediction.

1.4 Scope and Limitations

1.4.1 Scope

- Users can predict future house prices based on historical data and trends.
- Users can analyze the impact of various features such as location, area, BHK, etc., on house prices.
- Users can gain information into market trends and patterns, allowing them to make informed decisions.
- This system can be used anywhere at any time as it is a web-based application.

1.4.2 Limitation

- As users cannot achieve accurate predictions if the input data is incomplete, outdated or inaccurate.
- Users cannot expect the model to perfectly predict prices due to limitations in algorithms and potential overfitting or underfitting.
- Users cannot account for sudden market changes or external factors such as economic recessions, natural disasters) that are not represented in historical data.

1.5 Development Methodology

The project Real Estate House Price Prediction is best suited to waterfall approach as this methodology provides a structured, linear approach that is aligned with the clear and predictable stages needed in developing a house price prediction model. Since the project requires stages, such as gathering real estate data and selecting features which are well defined from the beginning. Waterfall methodology allows for a smooth development from data collection to model implementation and testing with minima changes. This process ensures that each phase is completed before going on to the next phase which minimizes risks and ensures a full review of the data and the accuracy of the model.

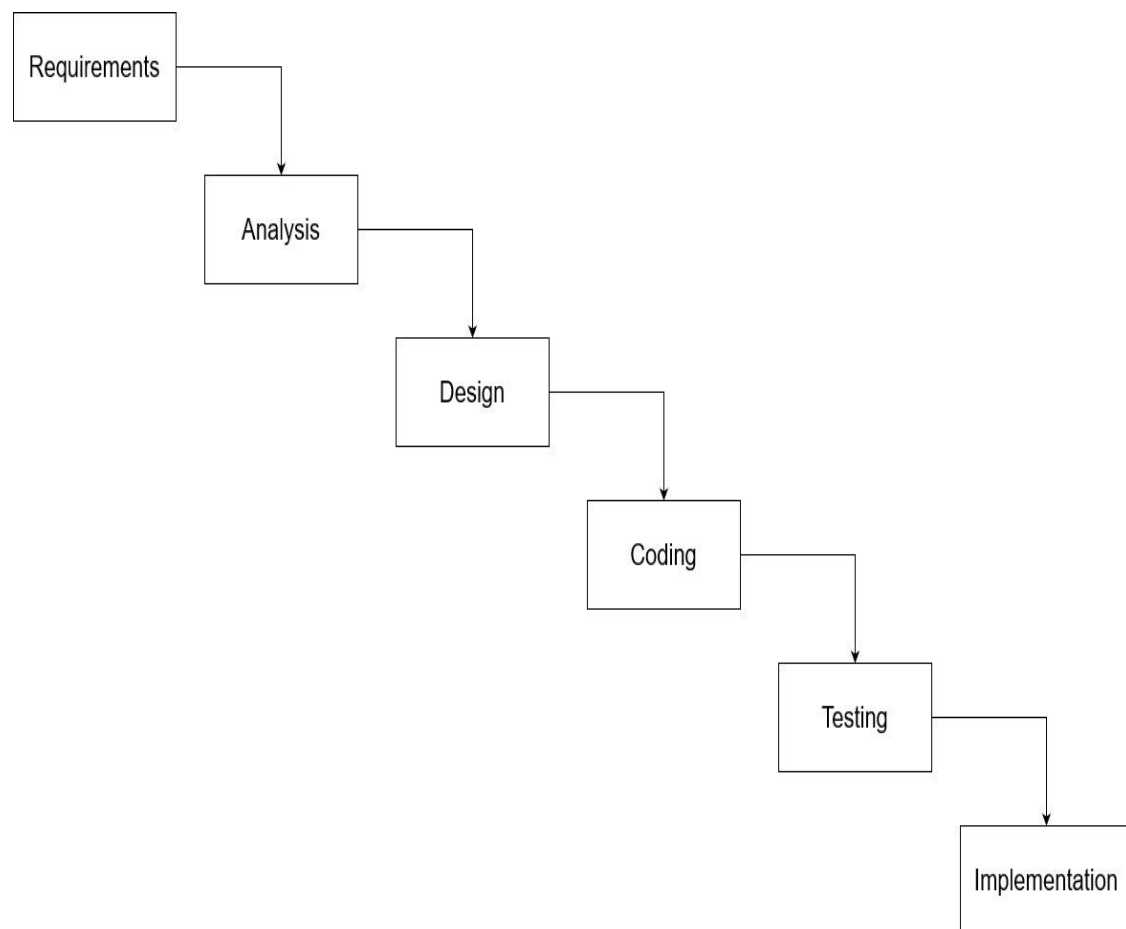


Figure 1. 1 Waterfall Methodology of Real Estate House Price Prediction

From the above diagram we can see that how waterfall's linear nature helps this project by dividing down the process into manageable phases: requirements gathering, analyzing specifications, designing required system architectures, coding, testing, implementing and maintaining the system by updating time to time. By following these phases, we will be ensured that each phase is thoroughly handled, lowering the risk of errors during the model creation and making the project more organized and predictable.

1.6 Report Organization

Chapter 1: Introduction

This chapter deals with the introduction of the system with its objectives and limitations along with the reason why the system is made. In addition, this chapter also cover the scope and limitation of the system.

Chapter 2: Background Study and Literature Review

Under this chapter, the information regarding the background of ecommerce system along with their system are covered. Likewise, the review of existing system similar to Ecommerce are carried out to find out flaws which are included here.

Chapter 3: System Analysis and Design

Here, all the system analysis and design component are included. Different software requirements i.e., functional and non-functional requirement and various feasibility study are also encompassed in this system analysis part. In addition, class: object diagram, sequence diagram, activity diagram and flowchart are also included.

Chapter 4: Implementation and Testing

Different tools used for implementing the system like case tools, programming language, database platforms are included in this chapter. Similarly, testing of different module and testing as a software are also included here. Result analysis of multiple models with their performance scores, along with visualizations to make results easier to understand are also included in this chapter.

Chapter 5: Conclusion and Future Recommendation

This chapter contains the lesson learned in the system development process, system outcome and conclusion are noted here. Likewise, the future recommendation and enhancement part from the developer perspective are also listed under this chapter.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Trends in housing prices indicate the current economic situation and also are a concern to the buyers and sellers. Many factors have an impact on house prices, such as the number of the bedrooms and bathrooms. House price depends upon its location as well. A house with great accessibility to highways, schools, malls and employment opportunities, would have a greater price as compared to a house with no such accessibility. Predicting house prices manually is a difficult task and generally not very accurate, hence there are many systems developed for house price prediction.

Real estate price study has been of great interest in research frontier. Discovering reliable and credible price indexes to track the rate of price appreciation over time and seizing the market trend have proven enormously beneficial for real estate professionals highly sought-after finance and business applications. Simply, price indexes could establish barometer of market information on risk and its turning point is paramount important to investment decision making, e.g., risk and return modelling, assets portfolio allocation strategy mapped price bubbles identification and evaluating investment performance.

2.2 Literature Review

A literature review is a comprehensive summary of previous research on topic by authorized scholars and researchers regarding of what has been done before on topic. The purpose of the literature review is to express and enumerate, summarized what ideas have been established and what their positive and negative aspects. There are many projects done on real estate house price prediction system some of them are as follows:

Real estate price prediction with regression and classification a project done on 2016 to predict house prices using dataset from Ames Iowa, comprising 1460 houses and 76 features. For regression model, the baseline linear regression model achieved a root mean square error (RMSE) of 0.5501. Lasso regression improved the RMSE to 0.5418 through feature selection and regularization, while Ridge regression has slight improvement with an RMSE of 0.5448. Support Vector Regression (SVR) with a Gaussian Kernel achieved RMSE of 0.5271, outperforming the baseline, whereas the SVR with a linear kernel performed poorly with an RMSE of 5.503. For classification models, Gaussian Naïve Bayes achieved an accuracy of 42% and Multi-nominal Naïve Bayes achieved 51%. Multi-nominal Logistic Regression had an accuracy of 50%, while Support Vector Machine

Classification (SVC) with a linear kernel reached 63% accuracy, and the Gaussian kernel achieved 41%. The Random Forest Classification model exhibited the highest accuracy at 67%. This study demonstrates that while traditional models provides a baseline, advance models such as SVR with a Gaussian kernel and Random Forest Classification offer significantly improved accuracy in predicting house prices. [1]

Gupta et al. 2024 explored the application of machine learning technique to predict real estate price in Mumbai focusing on the Decision Tree Regressor model. The study aimed to established a starting price for properties based on only geographical variables. The Decision Tree Regressor Model demonstrated high effectiveness with an accuracy of 89%, showcasing its potential in providing reliable market value predictions. This research emphasizes the utility of machine learning in enhancing accuracy and efficiency in the real estate market, helping clients make informed investment decision without relying on traditional brokers. [2]

House price prediction is a key supervise learning task, used to evaluate various machine learning models. This study compares the performance of three regression, using a dataset of sold houses. The LASSO regression model proved the most accurate, achieving 90.15% accuracy with a λ value of 0.001 and error rate of 0.0123. Linear and ridge regression also performed well, with accuracies close to 90%. Despite the small and simple dataset and the limited number of models evaluated, this study highlights the effectiveness of the LASSO regression model in predicting house prices and provides a foundation for future research to explore more complex datasets and advanced models. [3]

Predicting house sale price using machine learning is influenced by factors such as property area, location, construction material, property age, number of bedrooms and garages. This study evaluates several machines learning algorithm, including logistic regression, support vector regression (SVR), LASSO regression and decision trees. The R-squared values for logistic regression, SVR, LASSO regression and decision tree were 0.98, 0.96, 0.81 and 0.99, respectively. Performance was further assessed using MAE, MSE, RMSE and accuracy [4].

Predicting house price is critical task for real estate investors and planners, and machine learning algorithms have emerged as powerful tools for this purpose. Two notable algorithms in this domain are MLR (Multiple Linear Regression) and RF (Random Forest). MLR, which assumes a linear relationship between house prices and various features,

achieved and R^2 score of 0.73 in predicting house prices. This score indicates that MLR explains 73% of the variance in house prices. While MLR is effective in capturing linear trends, it struggles with more complex, non-linear relationships, which sometimes leads to prediction errors.

On the other hand, the random forest model, though it obtained a slightly lower R^2 score of 0.69, demonstrated superior performance in modeling intricate and non-linear relationships between house prices and features. RF excels in identifying important factors influencing house prices, such as house size, number of bathrooms, number of floors, parking spaces, and air conditioning. Its ability to handle complex data interactions makes it a valuable tool for more nuanced predictions. [5]

In real estate, predicting property prices accurately is crucial to avoid costly errors. This review tested different regression models to estimate property prices based on features like valuation date, road width, and location. The Multiple Linear Regression (MLR) model explained 60.41% of price differences and had a moderate error rate. The weighted Least Squares Regression (WLR) did better, explaining 65.01% of the differences with fewer errors. Penalized Linear Regression (PLR) had a high error rate despite explaining 62.08% of the differences, while MLR Gradient Descent showed slightly better error handling. Overall, WLR was the most accurate and reliable model for predicting commercial property prices [6].

Predicting house prices accurately is essential for making informed real estate decisions. This project utilized Linear Regression to forecast house prices based on features such as location, size, total square footage, number of bathrooms, and prices. The model achieved a high R^2 score of 0.761, indicating that approximately 76.1% of the variances in house prices was explained by the predictors. The Mean Absolute Error (MAE) for the Linear Regression Model was 17.576, suggesting that on average, the predicted prices were 17.576 units away from the actual prices. The Mean Absolute Percentage Error (MAPE) was 0.213, meaning the model's predictions were about 21.3% off from the true values on average. These metrics highlight the model's effectiveness and provide insights into accuracy and potential areas for improvement. [7]

Since the 1990s, China's real estate market has expanded rapidly, becoming a crucial part of the national economy. Accurate house price prediction is essential for effective regulation and investment strategies. This study developed a house price prediction model using

Multiple Linear Regression (MLR) and the Least Square Method. The model analyzed key factors such as residents' income, housing expenditure, and the completed real estate area. The results demonstrated high accuracy, with the model achieving a maximum prediction error of 8%. Specifically, errors range from as low as -0.2% to as high as 7.6% across different cities and years. The study highlighted significant positive correlations between house prices and residents' monthly income per capita disposable income, and per capita housing expenditure, while the completed real estate area had a negative correlation with prices. The MLR model effectively predicted real estate prices and provided valuable insights for market analysis, though future research could explore the impact of national policies on housing price. [8]

India's real estate market has surged due to its booming high-tech sector, favorable climate, and growing immigrant population, increasing the demand for property. This study developed a house price prediction model using machine learning to address demand. A dataset from Kaggle, encompassing variables like area, rooms, location, and amenities, was used to train and test various models. Among the models evaluated, Linear Regression performed the best with an accuracy of 84.5%, while Support Vector Regression (SVR) model showed the lowest accuracy at 64.5%. The results underscore the importance of selecting appropriate algorithms for accurate property price predictions. [9]

Accurate prediction of housing prices is crucial for making informed real estate decision, guiding investments, and reflecting economic trends. This study explores house price forecasting using three machine learning algorithms: Linear Regression, Random Forest and Gradient Boosting, applied to the Melbourne real estate dataset with 34,857 property sales and 21 features. The performance of the models was evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 Score. The linear regression model had an R^2 score of 0.822 and RMSE of 261,116.31 and a MAE of 171,486.27. The gradient boosting model performed best, with an R^2 score of 0.8282, an RMSE of 257,062.44 and a MAE of 169,009.36. These results indicate that Gradient Boosting provides superior accuracy compared to the other models. [10]

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

In the process of system designing many steps were taken into considerations. Among those steps, analyzing the system was one of the major steps. Different functional requirement, non-functional requirement discussion and feasibility study was implemented as a part of system analysis.

3.1.1 Requirement Analysis

The process of identifying the requirements that users must meet for a new project or system is known as requirement analysis. Organizations can identify the genuine demands of their stakeholders with the use of requirement analysis. The development team may converse with stakeholders in a language they can comprehend at the same time. For both consumers and developers, the Software Requirement Specification (SRS) document is simple to read. The following requirements are considered before beginning the creation of the new system.

i. Functional Requirement

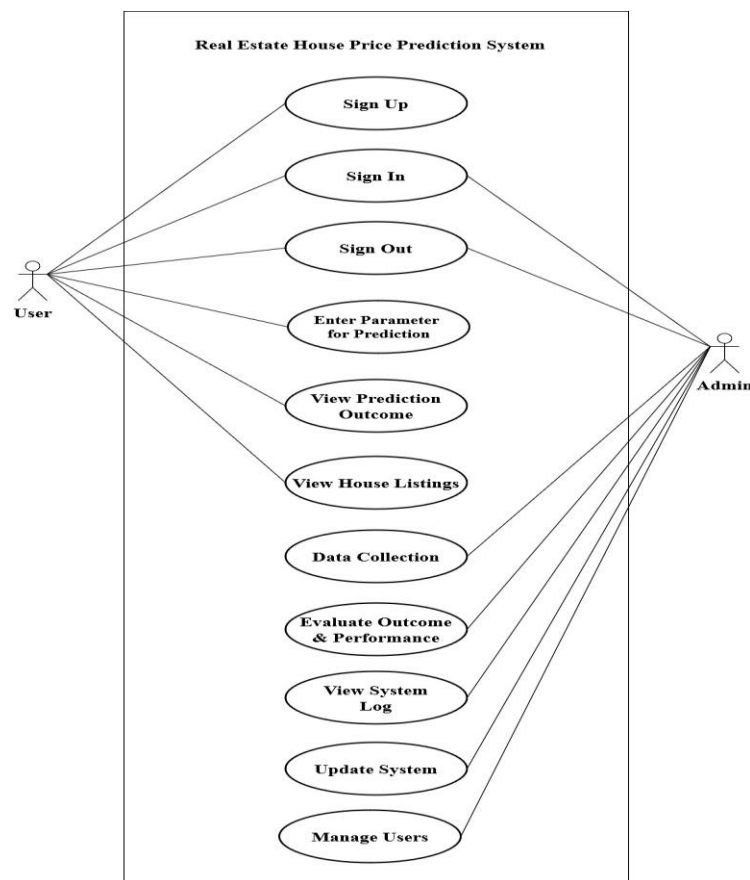


Figure 3. 1 Use Case Diagram of Real Estate House Price Prediction

The use case diagram above shows the interaction of the system with the external actors. Two actors are defined in the diagram: User and administrator. The user is the primary actor which interacts with the system by performing tasks like viewing predicted price, viewing house details, get recommended house lists, etc., The admin is another primary actor who interacts with the system by performing system level task like viewing system log, evaluating prediction outcome, updating the system and managing the users. The admin also collects data from various sources. The diagram helps to make it easier to understand how the system works and how different users can interact with it.

ii. Non- Functional Requirement

System attributes such as security, performance, reliability, etc. are defined by non-functional requirements. They ensure the whole system's performance and usability. If any one of them is not met, the system may not be able to meet the needs of the user. The following is a list of some of the non-functional requirements:

- **Performance:** The system provides fast response times, ensuring that the results load quickly and price predictions are mostly accurate.
- **Usability:** The system is easy to use and navigate, with a friendly interface that allows users to predict the price of the houses easily.
- **Reliability:** The system is reliable, available, and accessible 24/7.
- **Security:** The system is secure as it uses data encryption and passwords protection.
- **Maintainability:** As the code is written in a clear and readable format, making the system simple to maintain and update in the future.

3.1.2 Feasibility Study

A feasibility study evaluates and examines a project or system that has already been started. Several studies were carried out for the feasibility analysis. Here is a list of some of them:

i. Technical Feasibility

This system is technically feasible because it is simple to obtain the technical needed to put ideas into practice or convert system proposals into working software. As this project is created using an open-source and free tool, the hardware and software used to create it is at the cutting edge of technology. Therefore, there is not any technological challenges while developing this system.

ii. Operational Feasibility

This system is operational in part because it gives the company cost-effective information. Due to its user-friendly layout and the fact that accessing it just requires a user to have a basic understanding of computers and the internet

iii. Economic Feasibility

Since this system uses open-source development tools and doesn't need any additional software or hardware, there won't be any financial obstacles in completing the project. This system is both economically viable and cost-effective.

iv. Time Feasibility

The below Gantt chart represents the overall timeline of the project. It represents the sequential breakdown of the tasks involved in the project with the time taken for each task.

Table 3. 1 Gantt Chart of Real Estate House Price Prediction

Weeks Tasks	2081/03/25	2081/04/02	2081/04/10	2081/04/17	2081/04/24	2081/05/02	2081/05/10	2081/05/19	2081/05/26	2081/06/03	2081/06/07	2081/06/13
Requirements												
Analysis												
Design												
Coding												
Testing												
Implementation												
Documentation												

3.1.3 Object Modelling: Class & Object Diagram

The structure and behaviour of the proposed system is shown in the UML class diagram below. There are 5 classes present in the diagram they are User, Admin, Datasets, Prediction Model and Houses classes. Each class contains characteristics and behaviours defined by methods and attributes. The diagram also demonstrates how the classes are connected to one another by various forms of relations, which are represented by various lines. The relationship types employed in this diagram includes inheritance and aggregation. The admin class inherits or extends methods and attributes from user class. User and Admin class shares userId, name, email, password attributes and register, login

and logout methods. But admin class have unique is_admin attribute and manage users, update model, update datasets methods. There are four classes that have aggregation relation they are: Admin, Datasets, Prediction model and Houses. This relation represents that one class is part of another class but exists independently. There is many to many relations between dataset and prediction model because multiple dataset can be used to train multiple prediction model. The same relation is shared by user and houses because many users can query for multiple houses.

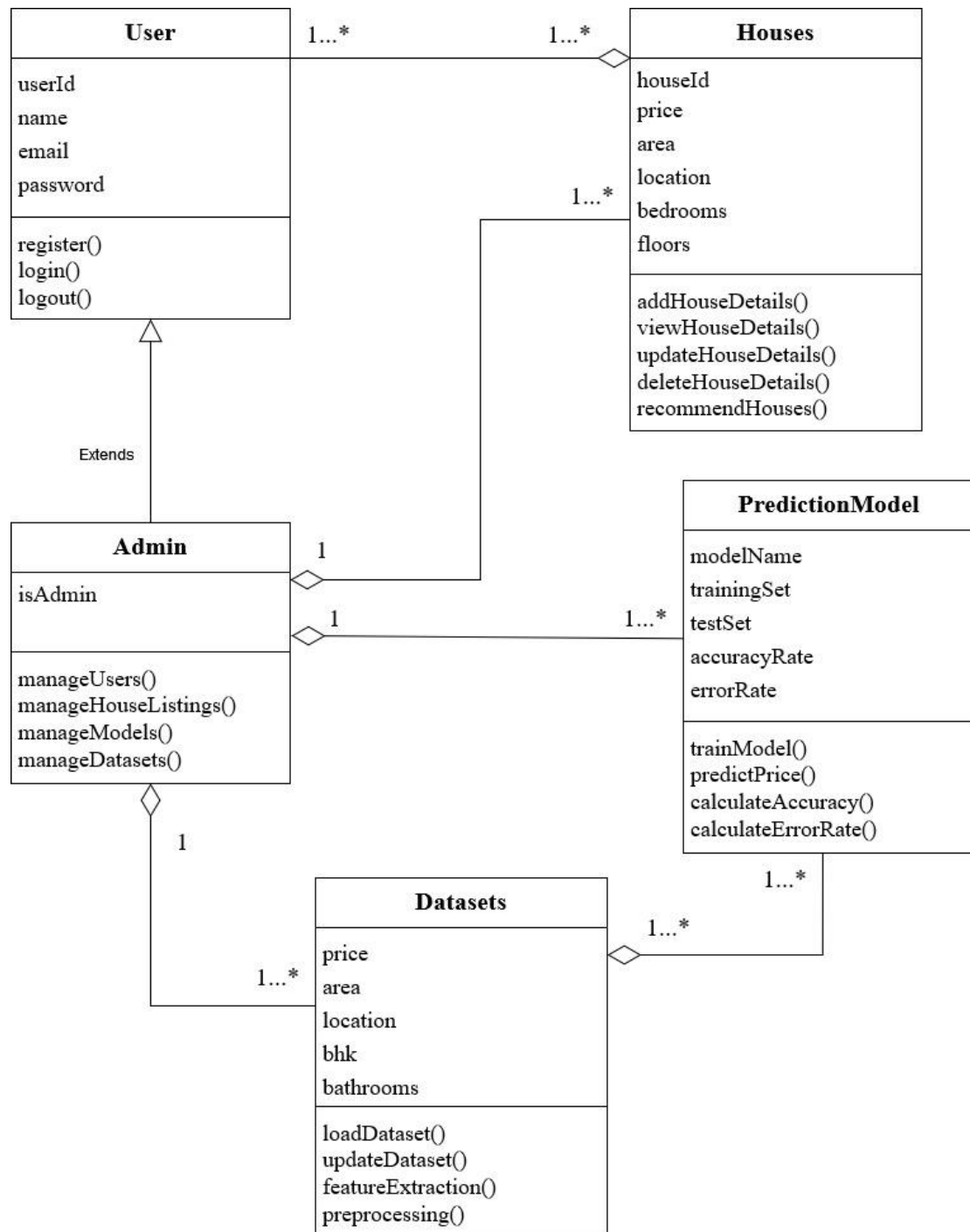


Figure 3. 2 Class Diagram of Real Estate House Price Prediction

The diagram below is the object diagram of real estate house price prediction, which represents the relationship between different classes in a system. As there are mainly five classes: user, admin, dataset, prediction model and houses. The attributes of each classes are labelled in the diagram below. For example, user class has attributes of userId, name, email etc., For example, the admin class is related with prediction model with one or more than one relationship and also aggregation relationship, which means that these classes are related with each other but can exists independently.

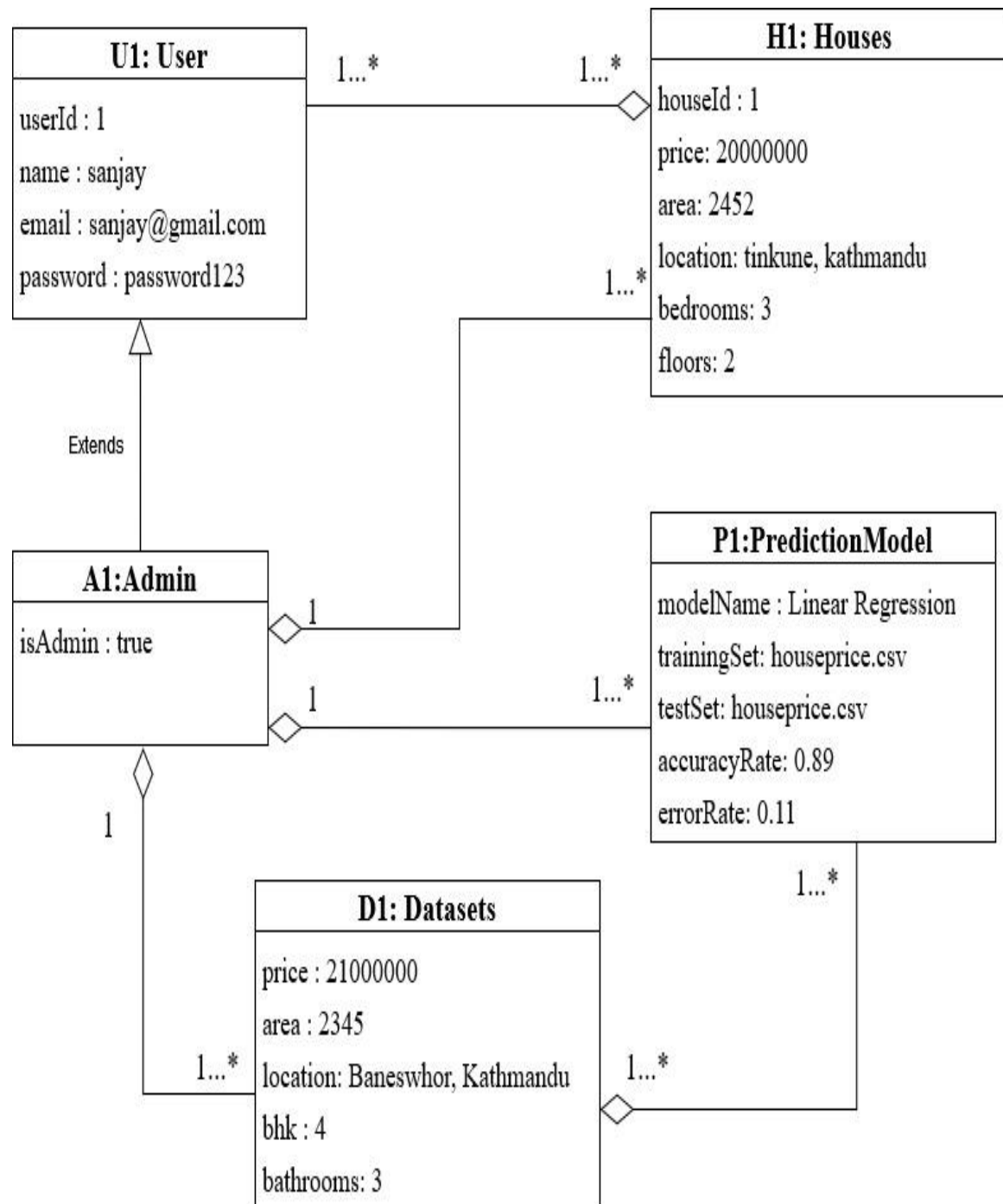


Figure 3. 3 Object Diagram of Real Estate House Price Prediction

3.1.4 Dynamic Modelling: State and Sequence Diagram

State Diagram:

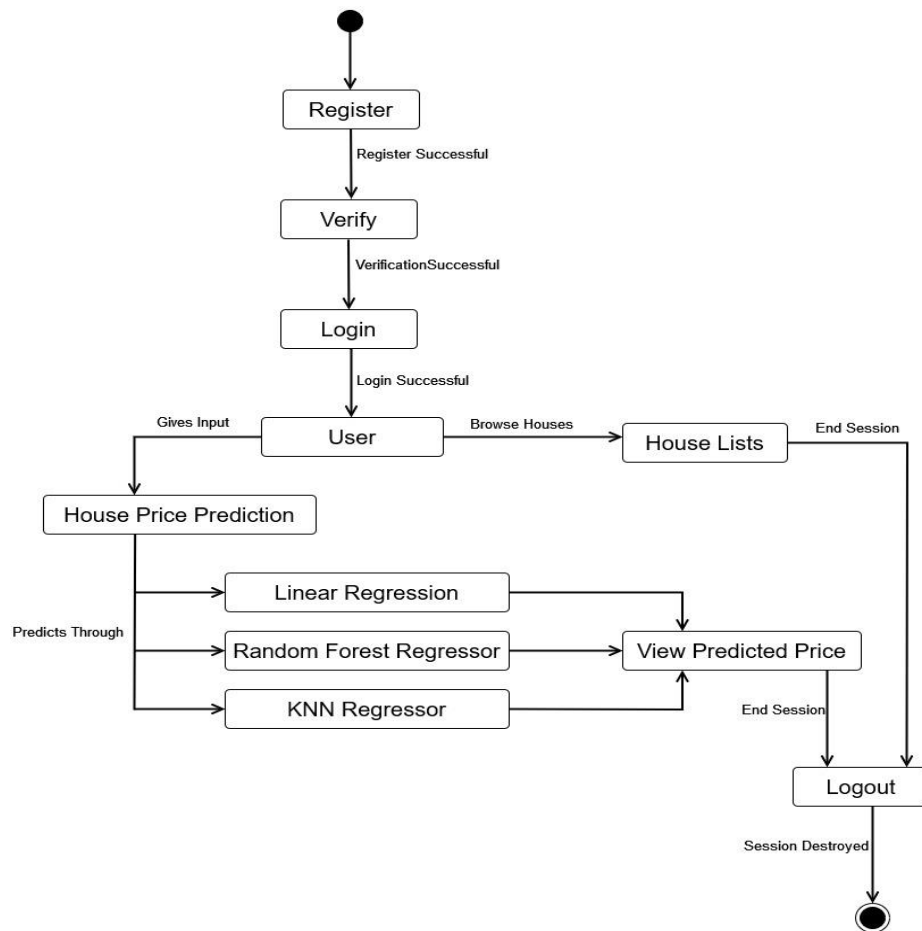


Figure 3. 4 State Diagram of Real Estate House Price Prediction

The above state diagram illustrates the flow of a user's interaction with a system that involves registration, login, browse house lists, predict house prices, and logout functionalities. The workflows begin at the initial state represented by a solid black dot, where user enters the required information to register. When the registration process is completed successfully, as shown by the condition "Registration complete" the user is taken to the verification state.

In the verification state the user enters their email and OTP (one-time-password) that is sent to their email. If the verification is successful, denoted by the condition "Verification Successful" the user is taken to the login state. In the login state, the user enters their credentials. If the login is successful, as shown by the condition "Login Successful", the user then moves to user state. The user state provides two options: the user can browse available properties through the house lists state, or make house price prediction through the house price prediction state.

If the user selects the house list state by triggering the “Browse House” condition, they are able to explore the available houses. When they decide to end browsing session, the condition “End Session” is triggered and moves to the logout state. If the user selects the predict house price state, they must enter the required information to predict the house price.

Within the house price prediction state, the process branches into three machine learning models: Linear Regression, Random Forest Regressor, and KNN Regressor. Each model processes the user’s input to generate a prediction. Once all the model has delivered their results, the system moves to view predicted price state, and values are displayed to the user. From both the house lists state and the view predicted price state, the session moves toward the logout state through the condition end session. Finally, the logout state leads to the session destruction, which moves to the terminal state represented by encircled black dot.

Sequence Diagram:

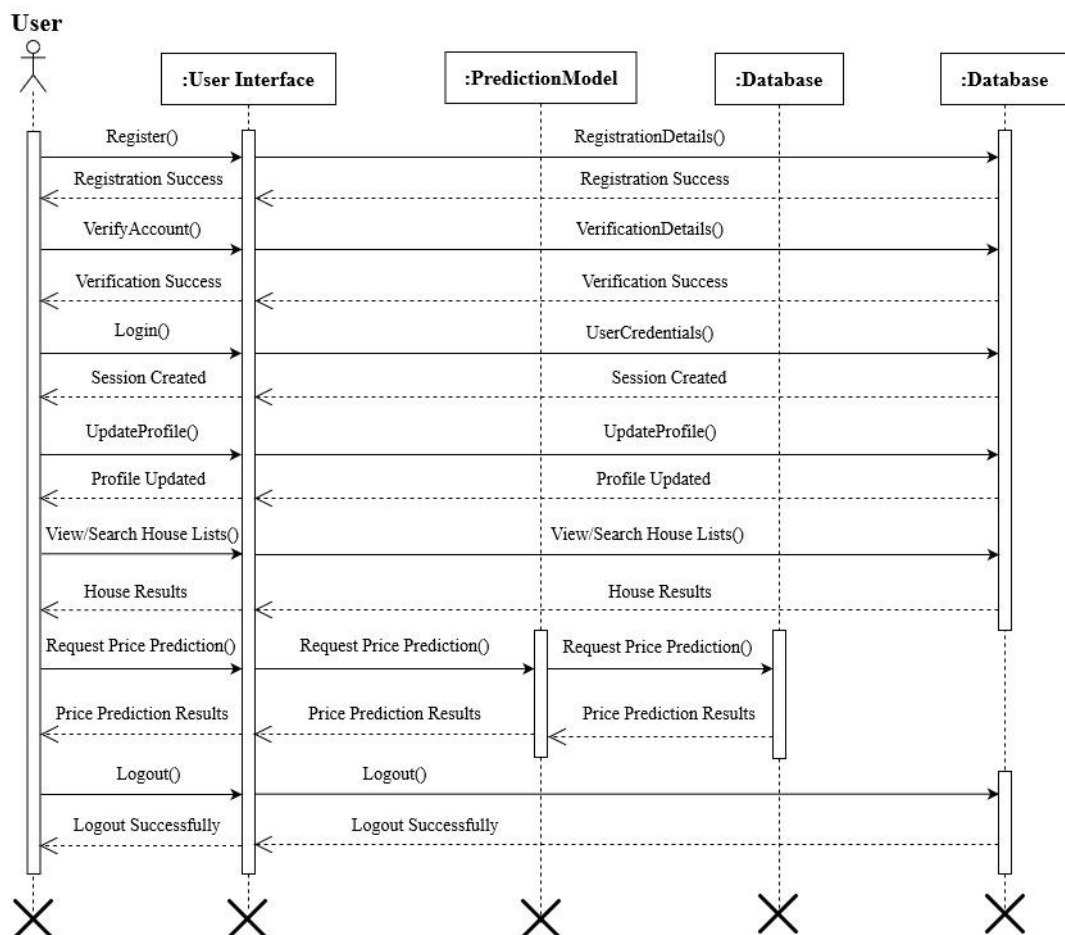


Figure 3. 5 Sequence Diagram of Real Estate House Price Prediction - User

The above sequence diagram of Real Estate House Price Prediction system below shows how user interacts with the system. This sequence diagram captures the flow of interaction and the order in which they occur, providing a clear understanding of the system's behaviour. There are mainly four objects in the user's side and admin's side: Users, Houses, Prediction model and Datasets. The admin and user actor can interact with these objects in the system. In database object user can store their login credentials and admin can manage users, houses information. The houses information later can be accessed from database by actor user later when browsing house lists. The first diagram is the interaction between actor user and object users, the user can register to the system by sending the message named register and the registration details, which will send reply message to the user's object. The users object sends the registration details to the database for storing the user data. The system validates and verifies the registration details and sends a reply with the registration success. After successful registration the verification email will be sent to the user's email with an OTP code. Then the user can verify their account by sending verify account and verification details message. After entering correct verification details the verification success message will be sent to user. Then the user can login to the system by sending message with the name of login and login credentials.

The second diagram of user side represents the interaction between user actor and houses objects. To get list of houses user can request in the system with get house lists to the houses object. The houses object then fetch the list of houses and sends house list as a message to the user. The user can also see house details by sending get house details message, which will reply with house details to the user. The user can also search for house list by sending search house, then house object will reply with the queried house lists.

The third diagram of user represents the prediction related actions. Only authenticated can use this feature. Authenticated users can request for price prediction and parameters of price prediction will be sent to both datasets and house objects. From the dataset the prediction model can predict the house price based on training and testing. From house objects, based on prediction parameters similar houses are recommended to the user.

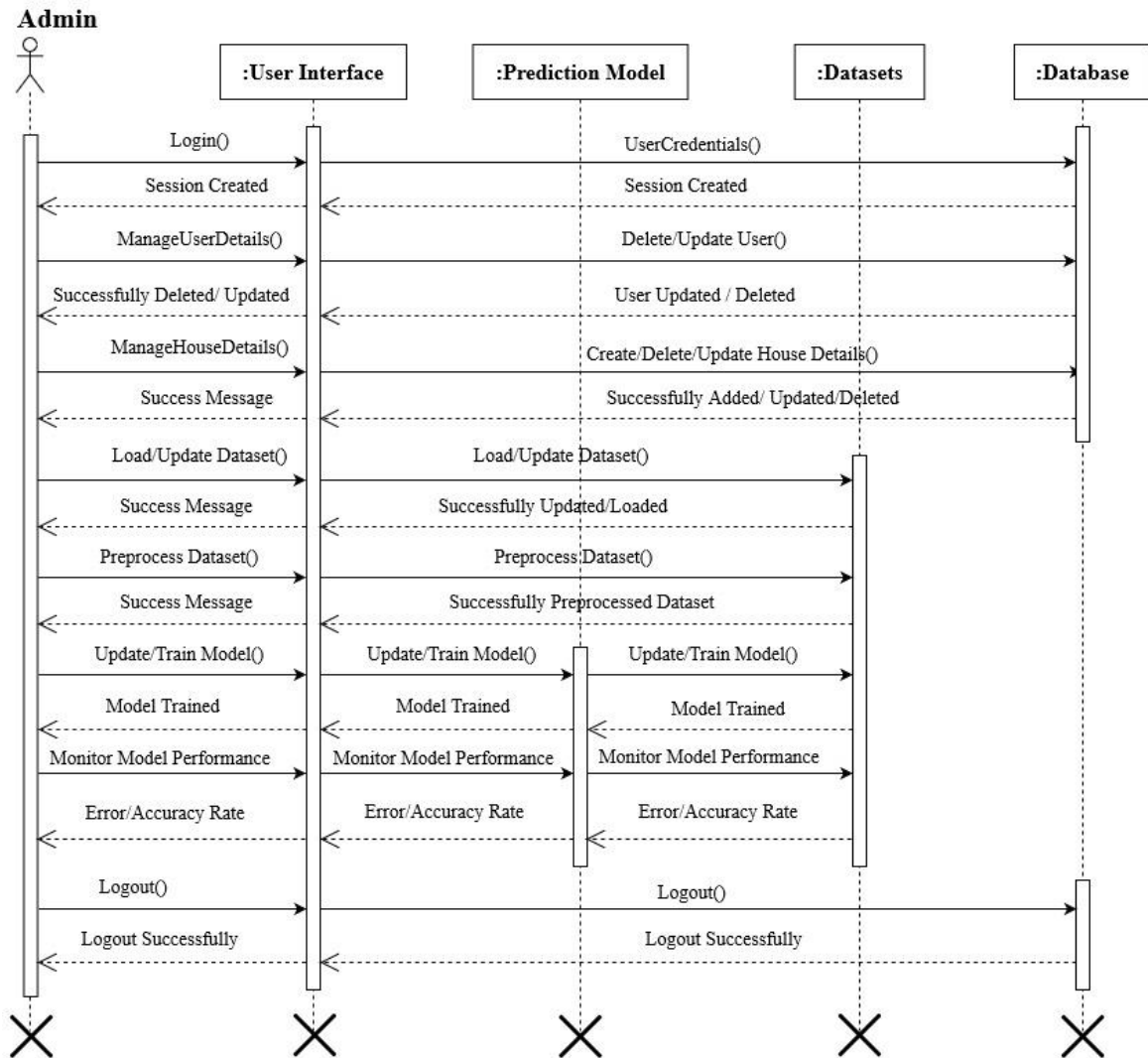


Figure 3. 6 Sequence Diagram of Real Estate House Price Prediction - Admin

The above diagram is the sequence diagram of the admin's side. The diagram represents every action performed by the admin must be also within the system. The first diagram is the interaction between actor admin and object users. The admin can login to the system by sending message with the name of login and login credentials in order to access admin dashboard. The admin and users have same process of login to the system. Then admin can get all user details by sending get user details message to the user's object. After getting user details admin can decide whether to update the user role by sending update user role message to the user's object which will update user's role in the database. The reply message user role updated will be returned to the admin. The admin can also delete the users by sending delete user message, which will then reply with user deleted message. The second diagram of user side represents the interaction between user admin and houses objects. To add new house details, add house details message must be sent to house object and it will store house details in the database and after that success message will be replied

from the houses object. The admin can also update house details by sending update house details, after success house details updated message will be sent to the admin. The house details can be deleted by sending delete house details message. Then house details deleted message will be replied after successful deletion.

The third diagram of user represents the training the prediction model related actions. The admin interacts with both the dataset and prediction model in this diagram. First the admin needs to load the data by sending load dataset message, which will reply as dataset loaded message. The admin can also update the old dataset by sending update dataset message. Different pre-processing techniques are applied by sending data pre-processing message in the dataset object. After successful pre-processing data pre-processing success will be returned as message to the admin. Then the pre-processed data can be used to train the model. The train model message is sent to the prediction model object which will also send message to dataset to fetch pre-processed datasets. After successful training model trained message will be sent back to the admin. The admin then can check the error rate and accuracy rate by sending calculate error and calculate accuracy message to the system. The system then replies with the error rate and accuracy rate. Later on, the model can be updated by sending update model message. The model replies with model trained message after successful update.

3.1.5 Process Modelling: Activity Diagram

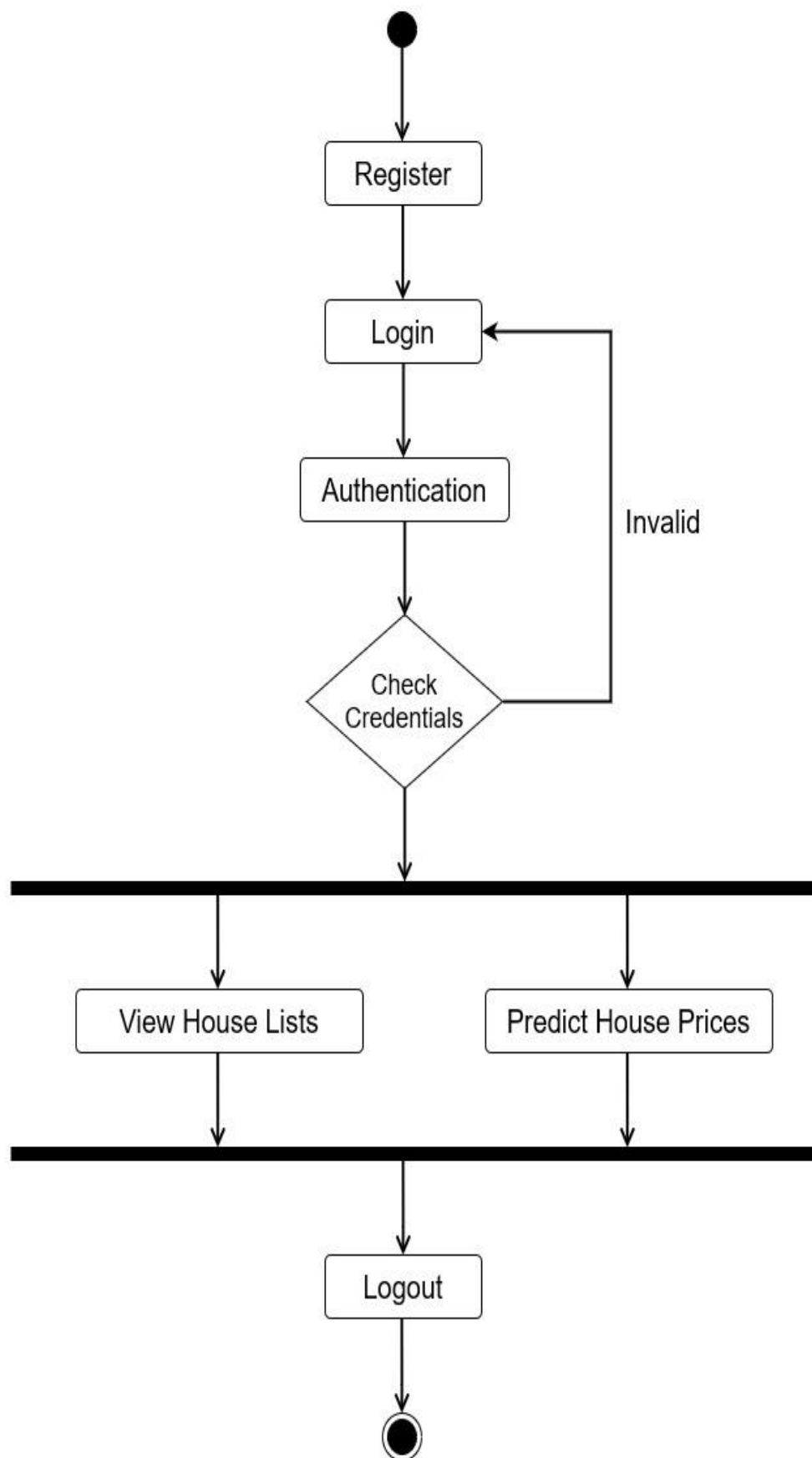


Figure 3.7 Activity Diagram of Real Estate House Price Prediction for user

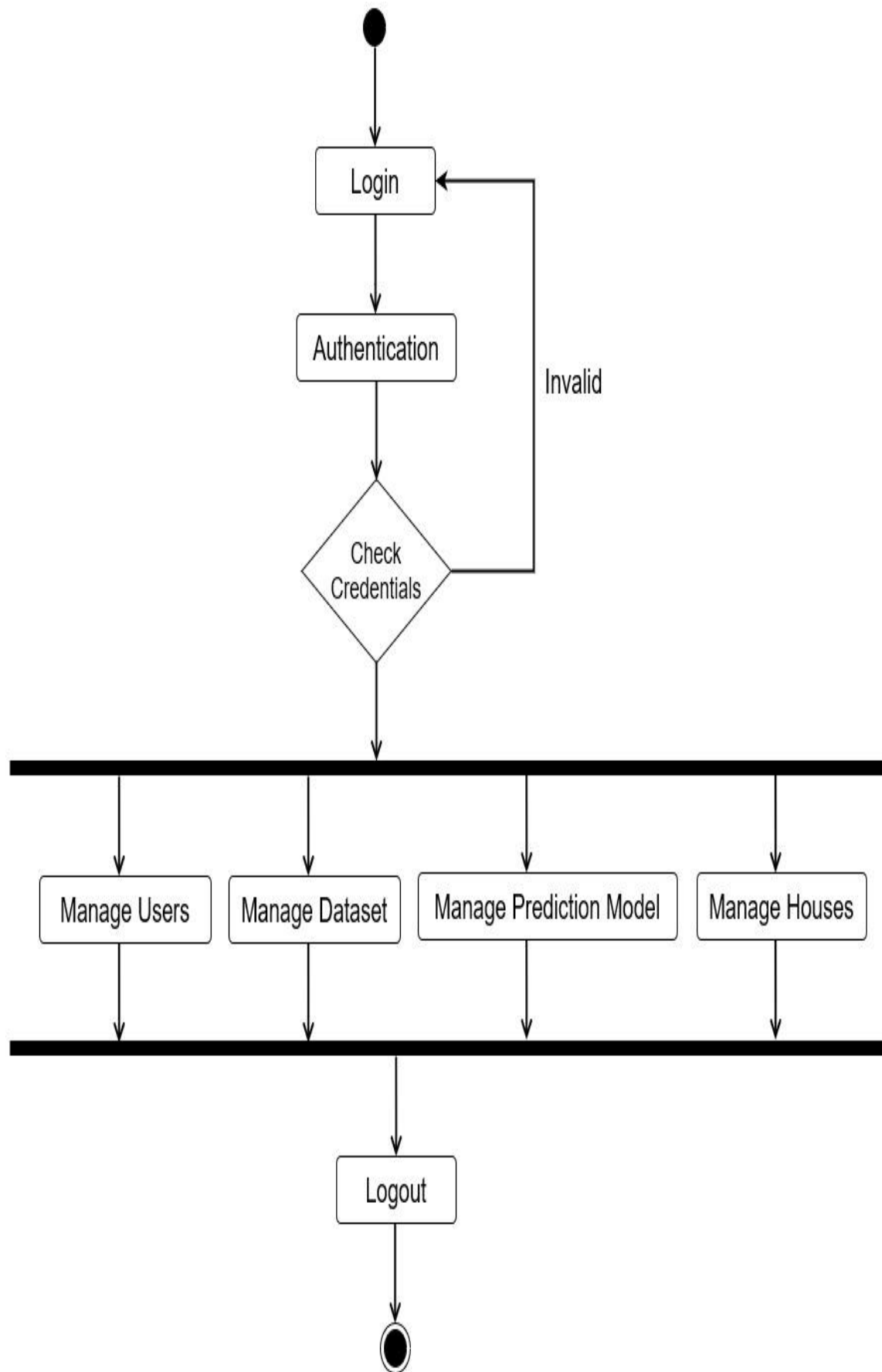


Figure 3.8 Activity Diagram of Real Estate House Price Prediction for admin

The activity diagrams above represent the process from sign up, sign in to sign out in the real estate house price prediction. The process starts with the black dot that represents the initial state or the start point of the activity. For the user, register is the first process after the application has started. This process is represented by rectangle with rounded corners. After that leads to the login process where all the checking of credentials performed. This login process is same for admin also. If the credentials are correct then both the admin and user are redirected to their respective dashboards. If the credentials are incorrect then the are redirected to login process again. The user can either view house lists or predict price of the house. Likewise, after entering in admin dashboard, the admin can manage users, datasets, prediction models or houses. The admin can update user role or delete users from manage users. Admin can also manage house lists by adding, deleting or updating them. Datasets can be managed by updating them or removing the outliers. Similarly, prediction model can be manged by feeding more datasets, implementing more advanced model or ensemble learning to predict house prices more accurately.

3.2 System Design

3.2.1 Refinement of Classes and Objects

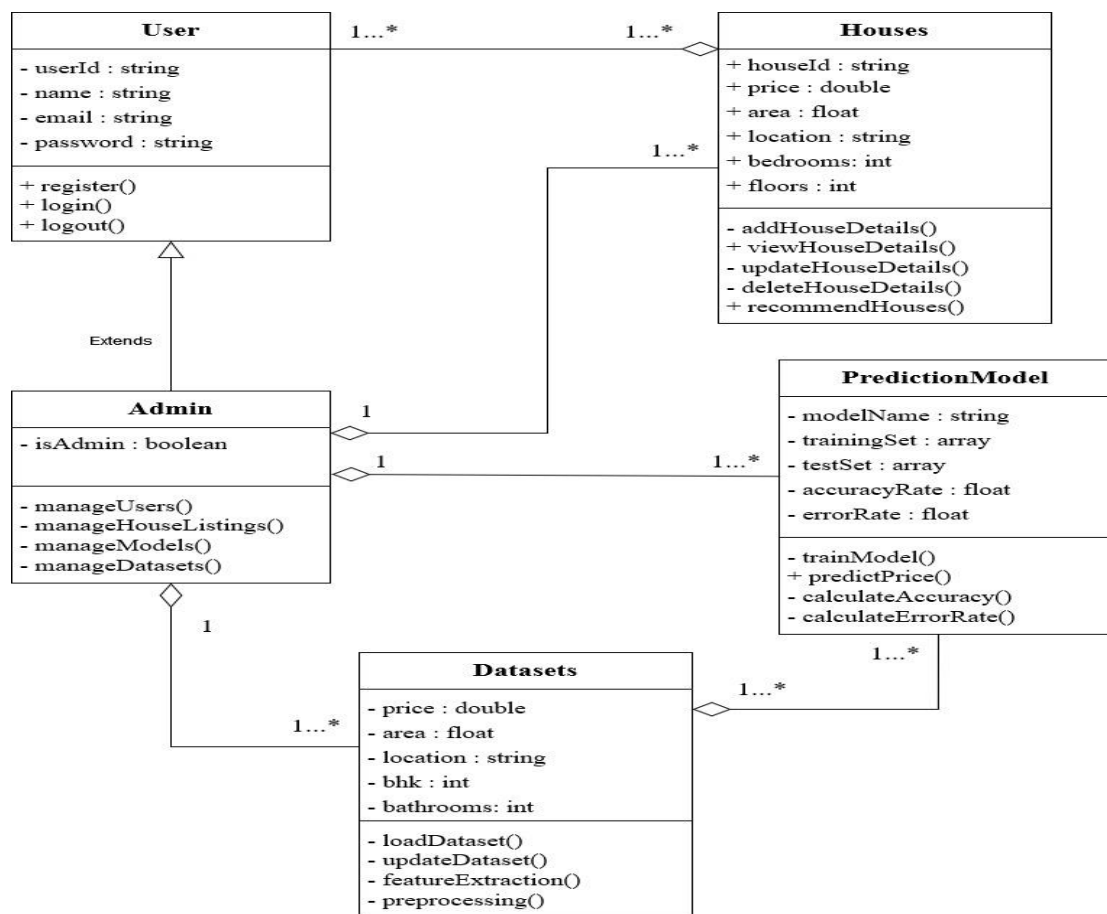


Figure 3.9 Refinement of Class Diagram of Real Estate House Price Prediction

In real estate house price prediction, refinement of class diagram involves making detailed adjustment or improvement to the diagram to provide more specific and precise information about the classes, their attributes, methods and relationships. In above diagram, there is a notation of visibility for each attribute and operations like public and private. In the system there are five classes named users, admin, prediction model, datasets and houses. The plus “+” sign denotes the public access modifier whereas, minus “-” sign denotes the private access modifier. The attributes in all class has private access modifier except houses class. The houses class have public access modifier. All the data types (String, int, Boolean, etc.,) of attributes present in the classes are also labelled in the diagram. The users class contains public access modifier login, register and logout methods. The view house details and recommend house of house class and predict price methods of prediction model class has public access modifier. All others methods have private access modifier (i.e., load dataset, manage dataset, add house, etc.,).

3.2.2 Component Diagram

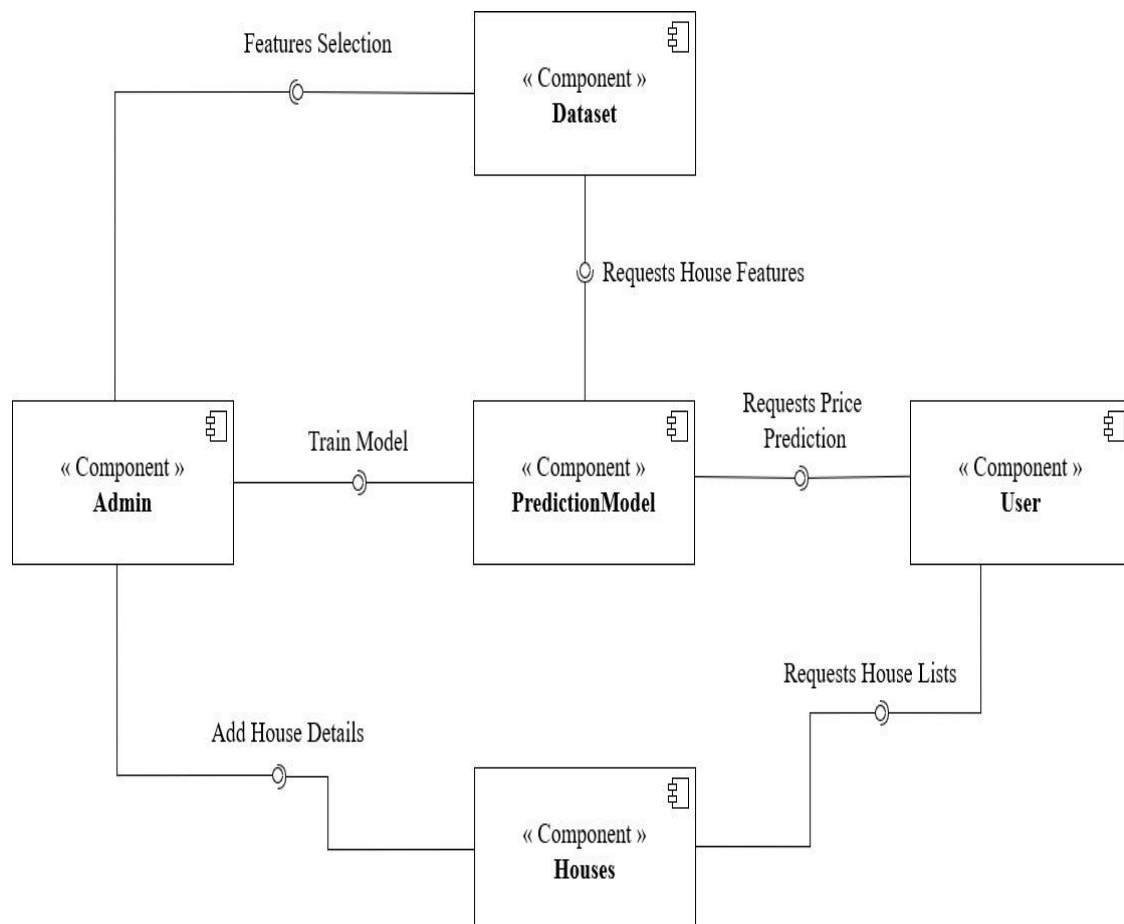


Figure 3. 100 Component Diagram of Real Estate House Price Prediction

The component diagram above shows the relationship between different components in Real estate house price prediction application. The diagram shows how each component

interact with each other to perform different actions such as adding house details, training the model, predicting the house prices, etc., In the diagram there are mainly five components they are: users, admin, prediction model, datasets and houses. The rectangle with small rectangle in the upper right corner with tabs represents the components. The semi-circle on a line represents a required interface and full circle with line represents provided interface.

From the above diagram we can see that component admin provides the details to add house details in the house component. The admin also provides different training models to perform the price prediction in the prediction model. The prediction model requires dataset to train and test from. The prediction model requests dataset from dataset component. The user also requests for price prediction based on the parameters they provide to the prediction model. The user can also request for the house list they want to browse. The user can provide different search query and filter query to request house lists.

3.2.3 Deployment Diagram

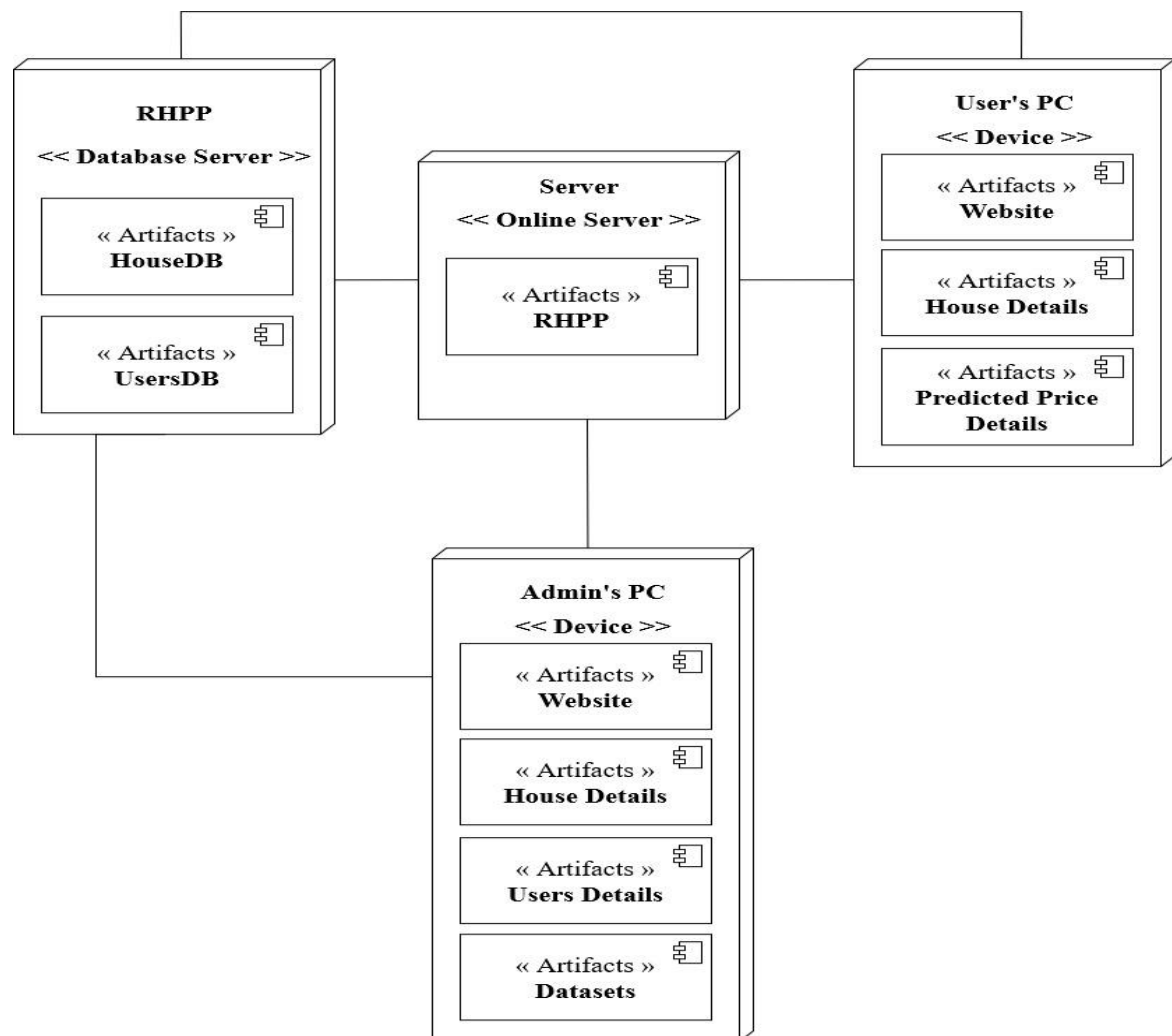


Figure 3. 111 Deployment Diagram of Real Estate House Price Prediction

The deployment diagram above represents the relationship between different nodes and components of the system. The system consists of four nodes: Real estate house price prediction server, database server, admin and user nodes. The database server node is responsible for storing users and houses information. The database server node is connected to real estate house price prediction server through the association relation. The server node is responsible for processing information provided by the users, such as authentication, searching house lists, predicting house prices, etc., The server node is connected to the database server node, user node and admin node with an association relation. This node also processes users, houses, and price prediction information provided by the admin. The admin node is responsible for showing the admin dashboard, which consist of processes to manage users, datasets, prediction model and houses. The admin node is connected to server node and database node with an association relationship. The user node is responsible for accessing to the system, from which the user can see house lists, predict price, get recommended houses lists, etc., The user node is connected to the database and server node through the association relationship.

3.3 Algorithm

The system uses Regression Based Algorithms for predicting the prices of the houses. Regression based algorithms is best suited for this system because they work well with continuous numerical values. Housing prices are numbers that can go up or down making them a perfect fit for regression models, which are designed to handle this type of data. These algorithms are good at finding relationships between different features of house, like the number of bedrooms, the size of the house, its location and the price. By understanding how these features interact with the price, regression models can make accurate predictions. Following algorithms are used in the project:

a. Multiple Linear Regression

Multiple linear regression is a fundamental component of the house price prediction system by providing a straightforward and understandable model that highlights the relationship between a feature such as: area (in sq. ft.), BHK and bathrooms, location and house prices. Its simplicity makes it easy to implement and understand, allowing stakeholders to quickly catch how certain features affect house prices.

How it works?

Step 1: Define the variable

- **Dependent variable (Target):** House price (the value to be predicted)

- **Independent variable (Features):** Location, area, BHK, and bathroom.

Step 2: Formulate the MLR equation

The MLR equation establishes a relationship between the dependent variable (price) and the independent variables (features). It can be expressed as:

$$y = m_0 + m_1 \cdot x_1 + m_2 \cdot x_2 + m_3 \cdot x_3 + m_4 \cdot x_4$$

where,

- y represents the house price.
- m₀ is the intercept, or base price when all features are zero.
- m₁, m₂, m₃, and m₄ are coefficients indicating the influence of each feature on the house price.
- x₁, x₂, x₃, and x₄ corresponds to location, area, BHK, and bathroom respectively.

Step 3: Train the model

- **Data pre-processing:** Remove outliers to enhance model's accuracy and apply one-hot encoding on categorical variables (location) to transform non-numeric data into a suitable format for analysis.
- **Model training:** Fit the MLR model using historical data to learn the coefficients that minimize the prediction error

Step 4: Make Predictions

- **Input features:** The user provides input characteristics for the house (location, area, BHK, and bathrooms).
- **Prediction calculation:** The trained model applies the equation to estimate the house price:

Step 5: Evaluate the model's performance

- **Use metric system:** To ensure the model's accuracy and adjust for improvements, performance is evaluated using r-squared, mean squared error, and root mean squared error.

b. Random Forest Algorithm

Random Forest improves the house price prediction system by implementing the power of ensemble learning. By building and averaging multiple decision trees, this algorithm handles complex, non-linear relationships and interactions between features.

How it works?

Step 1: Define the variable

- **Dependent variable (Target):** House price (the value to be predicted)

- **Independent variable (Features):** Location, area, BHK, and bathroom.

Step 2: Model overview

The random forest regressor is an ensemble learning method that builds multiple decision trees and combine their prediction to improve accuracy and reduce overfitting.

Step 3: Train the model

- **Data pre-processing:** Remove outliers to enhance model's accuracy and apply one-hot encoding on categorical variables (location) to convert them into a usable format for analysis.
- **Model training:** Train the random forest regressor on historical data by constructing multiple decision trees. Each tree is trained on a random subset of data and features to ensure diversity in predictions. The model combines the predictions from all trees, often using an average, to produce a final prediction for house price.

Step 4: Make Predictions

- **Input features:** The user provides input characteristics for the house (location, area, BHK, and bathrooms).
- **Prediction calculation:** The random forest model aggregates predictions from each decision tree in the forest to output a final estimated house price.

Step 5: Evaluate the model's performance

Use metric system: To ensure the model's accuracy and adjust for improvements, performance is evaluated using r-squared, mean squared error, and root mean squared error.

c. K-Nearest Neighbour (KNN) Algorithm

K-Nearest Neighbour adds value to the house price prediction system by providing a simple yet effective method for capturing local patterns in the data. By identifying the 'K' most similar data points and averaging their house prices. KNN makes accurate predictions based on local neighbourhoods of features.

How it works?

Step 1: Define the variable

- **Dependent variable (Target):** House price (the value to be predicted)
- **Independent variable (Features):** Location, area, BHK, and bathroom.

Step 2: Model overview

The K-Nearest Neighbour (KNN) regressor predicts the target variable based on the values of its closest neighbours in the feature space. It assumes that similar houses have similar prices.

Step 3: Train the model

- **Data pre-processing:** Remove outliers to enhance model's accuracy and apply one-hot encoding on categorical variables (location) to enable proper distance calculations.
- **Model training:** Select the number of neighbour K and train the model by storing historical data. During prediction, the KNN regressor calculates the distance between the target and its nearest k neighbours and uses this to predict the house price.

Step 4: Make Predictions

- **Input features:** The user provides input characteristics for the house (location, area, BHK, and bathrooms).
- **Prediction calculation:** The trained model identifies the k nearest neighbours of the input data and predicts the house price by averaging the prices of these neighbours

Step 5: Evaluate the model's performance

Use metric system: To ensure the model's accuracy and adjust for improvements, performance is evaluated using r-squared, mean squared error, and root mean squared error.

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

Designing, coding and developing this application is done by the help of different tools, frameworks and programming languages. This project is built with the frameworks and libraries of JavaScript and Python. These libraries and frameworks are used to simplify the creation of application and follows the model-view-controller (MVC). The following design and development tool made the process easier.

React JS: For the frontend development React JS is used. React JS is used for the frontend. The overall user interface (UI) is built using this library. Both admin and user side interface are built using this library. React JS is mostly used to design layout for the web platforms.

Context API & Axios: For the state management Context API is used, while communicating with backend axios is used. These tools are simple and straightforward tool that is used to handle state and communicate with backend server.

Node JS & Nest JS: Node JS is a runtime environment used to execute JavaScript code outside browser. It is mostly used for building backend services like APIs like website or applications, Nest JS is a Node JS framework for building efficient, scalable Node JS server-side applications. It uses TypeScript and combines elements of OOP (Object Oriented Programming).

PostgreSQL: PostgreSQL is a free open-source database system that supports both relational (SQL) and non-relational (JSON) queries. PostgreSQL is a back-end database for dynamic websites and web applications. PostgreSQL is an advanced relational database system.

Python: Python is used for machine learning due to its simplicity and the powerful libraries like Pandas, NumPy and Scikit-Learn. This language is used for building and training the machine learning model to predict the price of the house based on the features provided.

Flask: Flask is a lightweight python web framework that allows to build web servers for serving the machine learning models or APIs. It's often used for creating RESTful API to integrate python machine learning models into web applications.

4.1.2 Implementation Details of Modules

The system named “real estate house price prediction” uses different modules. A module is a software component or part of a program that contains one or more routines. The

implementation of these modules plays a crucial role overall functionality and success of the application. These modules are:

a. Registration Module: The registration module is responsible for creating new users for the system. This module requires details like name, email, password, phone numbers, etc., then account verification code is sent to the customer email address for verification process.

```
You, last week | I author (100) | Codeium: Refactor | Explain
@ApiTags('authentication')
@Controller('auth')
export class AuthController {
  constructor(private readonly authService: AuthService) {}

  // ----- REGISTER ----- //
  @Post('register')
  @HttpCode(HttpStatus.CREATED)
  @ApiConsumes('multipart/form-data')
  @ApiConsumes('application/json')
  @UseInterceptors(
    FileInterceptor('avatar', {
      storage: diskStorage({
        destination: 'static/avatar',
        filename: getFileName,
      }),
      fileFilter: imageValidator,
    }),
  )
  @ApiOperation({
    summary: 'Register a user',
    description: 'All users can register here',
  })
  register(
    @Body() payload: RegisterDTO,
    @UploadedFile() file?: Express.Multer.File,
  ) {
    return this.authService.register(payload, file);
  }
}
```

You, 2 weeks ago • auth module completed

Codeium: Refactor | Explain | Generate JSDoc | X

Figure 4. 1 Registration Module of Real Estate House Price Prediction

b. Login Module: The login module is responsible for authenticating the verified users. Users with correct credentials will be redirected to their respective dashboard. Only authenticated users can use prediction model and can get recommended house lists. Unauthenticated users can still browse the house listings and get house details, but they will not be authorized to perform actions like authenticated users.

```

export class AuthService {
  async login(payload: LoginDTO) {
    const { email, password } = payload;

    // check if user exists
    const user = await this.dataSource.getRepository(User).findOne({
      where: { email },
      select: [ ...
    ]},
    });

    if (!user)
      throw new BadRequestException({ ...
    });

    // validate password
    const isPasswordMatched = await argon.verify(user.password, password);

    if (!isPasswordMatched)
      throw new BadRequestException({ ...
    });

    // generate jwt token
    const token = await this.jwtService.signAsync( ...
    );

    return {
      token,
      role: user.user_type,
      user: { ...
    },
      message: `Welcome back, ${user?.name}`,
    };
  }
}

```

Figure 4. 2 Login Module of Real Estate House Price Prediction

c. Verification Module: This module is responsible for verifying the genuine user. If the user has provided correct email, then they will receive verification code. They can verify their account in this module. Only verified users can proceed for login otherwise they will not be able to access in their particular dashboard.

```

export class AuthController {
  // ----- VERIFY ----- //
  @ApiOperation({
    summary: 'Verify Email',
    description: 'User email verification',
  })
  @HttpCode(HttpStatus.OK)
  @ApiBearerAuth('JWT-auth')
  @UseGuards(JWTAuthGuard)
  @Post('email-verify')
  verify(@Body() payload: VerifyEmailDTO) {
    return this.authService.verifyEmail(payload);
  }
}

```

Figure 4. 3 Verification Module of Real Estate House Price Prediction

d. Houses Module: In this module list of available houses are displayed that are fetched from the database. Users can also search for the houses with different filters they like. They can also view details of individual house. Recommended houses are also fetched from this module during predicting the house prices.

```
export class PropertiesController {  
  // ----- GET ALL PROPERTIES- PUBLIC ROUTE ----- //  
  @Get()  
  @HttpCode(HttpStatus.OK)  
  @ApiOperation({  
    summary: 'Get all properties',  
    description: 'Get all properties',  
  })  
  getAllProperties(@Query() query: GetPropertiesQuery) {  
    return this.propertiesService.getAllProperties(query);  
  }  
  
  // ----- GET PROPERTY DETAIL - PUBLIC ROUTE----- //  
  @Get('/:slug')  
  @HttpCode(HttpStatus.OK)  
  @ApiOperation({  
    summary: 'Get property details',  
    description: 'Get all property details',  
  })  
  getPropertyDetails(@Param('slug') slug: string) {  
    return this.propertiesService.getPropertyDetails(slug);  
  }  
}
```

Figure 4. 4 Houses Module of Real Estate House Price Prediction

e. Dashboard Module: After successful authentication users will be redirected to their respective dashboard. Users will be redirected to their profile dashboard from where they can manage their profile. Users can also navigate to predict house price page to predict the price of house with different features. Admin will be redirected to the admin dashboard from where admin can manage users, houses, datasets and prediction model.

```

you, yesterday | I author (you)
@Controller('properties')
export class PropertiesController {
  constructor(private readonly propertiesService: PropertiesService) {}

  // ----- REGISTER A PROPERTY - ADMIN ROUTE ----- //
  @Post()
  @HttpCode(HttpStatus.CREATED)
  @ApiBearerAuth('JWT-auth')
  @UseGuards(JWTAuthGuard, RolesGuard)
  @ApiConsumes('multipart/form-data')
  @ApiConsumes('application/json')
  @Roles(UserType.ADMIN)
  @ApiOperation({
    summary: 'Register a new property',
    description: 'Register a new property',
  })
  @UseInterceptors(
    FilesInterceptor('image', 5, {
      storage: diskStorage({
        destination: 'static/properties',
        filename: getFileName,
      }),
      fileFilter: imageValidator,
    }),
  )
  registerProperty(
    @Body() payload: RegisterPropertyDTO,
    @UploadedFiles() files: Array<Express.Multer.File>,
  ) {
    return this.propertiesService.registerProperty(payload, files);
  }
}

```

Figure 4. 5 Dashboard Module of Real Estate House Price Prediction

f. Prediction Module: This module is responsible for predicting the prices of houses based on the parameters that authenticated users provides. This model fetches data from the dataset provided by the admin. The model is trained from the test set divided from the dataset and validates the model from validation set to prevent overfitting.

```

Codeium: Refactor | Explain
class MyLinearRegression:
    Codeium: Refactor | Explain | Generate Docstring | X
    def __init__(self, lr=0.001, n_iters=1000):
        self.lr = lr
        self.n_iters = n_iters
        self.weights = None
        self.bias = None

    Codeium: Refactor | Explain | Generate Docstring | X
    def fit(self, X, y):
        n_samples, n_features = X.shape
        self.weights = np.zeros(n_features)
        self.bias = 0

        for _ in range(self.n_iters):
            y_pred = np.dot(X, self.weights) + self.bias

            dw = (1/n_samples) * np.dot(X.T, (y_pred-y))
            db = (1/n_samples) * np.sum(y_pred-y)

            self.weights = self.weights - self.lr * dw
            self.bias = self.bias - self.lr * db

    Codeium: Refactor | Explain | Generate Docstring | X
    def predict(self, X):
        y_pred = np.dot(X, self.weights) + self.bias
        return y_pred

    Codeium: Refactor | Explain | Generate Docstring | X
    def score(self, X, y):
        X = X
        y = y
        y_predicted = self.predict(X)
        ss_total = np.sum((y - np.mean(y)) ** 2)
        ss_residual = np.sum((y - y_predicted) ** 2)
        r2_score = 1 - (ss_residual / ss_total)
        return r2_score

```

Figure 4. 6 Prediction Module of Real Estate House Price Prediction

g. Dataset Module: This module is responsible for providing test set, train set and validation set to train, test and validating the model. Different pre-processing like feature extraction, feature selection, dimensionality reduction is performed to make the dataset easy to interpret for the prediction model. To make the model more accurate the dataset must be real, correct and must not contain null, fake and outlier's data.

Features of Dataset:

	A	B	C	D	E	F	G	H	I
1	area_type	availability	location	size	society	total_sqft	bath	balcony	price
2	Super built-up Area	19-Dec	Electronic City Phase I	2 BHK	Coomee	1056	2	1	39.07
3	Plot Area	Ready To Move	Chikka Tirupathi	4 Bedroom	Theanmp	2600	5	3	120
4	Built-up Area	Ready To Move	Uttarahalli	3 BHK		1440	2	3	62
5	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	Soiewre	1521	3	1	95
6	Super built-up Area	Ready To Move	Kothanur	2 BHK		1200	2	1	51
7	Super built-up Area	Ready To Move	Whitefield	2 BHK	DuenaTa	1170	2	1	38
8	Super built-up Area	18-May	Old Airport Road	4 BHK	Jaades	2732	4		204
9	Super built-up Area	Ready To Move	Rajaji Nagar	4 BHK	Brway G	3300	4		600
10	Super built-up Area	Ready To Move	Marathahalli	3 BHK		1310	3	1	63.25

Figure 4. 7 Dataset Module of Real Estate House Price Prediction

The above diagram of house price datasets is sourced from Kaggle.com. Initially, the dataset had 9 features, which were later reduced to the following 5 key features: location, price, area, BHK, and bathrooms. Here's a brief description of each feature:

- **Location:** The specific area where the property is located.
- **Price:** Represents the selling price of the property.
- **Area:** Represents the total land area, including, house, garden, parking etc.,
- **BHK:** Represents the total number of bedrooms, along with a hall and kitchen.
- **Bathrooms:** Represents the count of bathrooms available in the property.

4.2 Testing

4.2.1 Test Case for Unit Testing

Table 4.1 Test Case for Starting the Server

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC1.1	Start client, backend and prediction model server in terminal.	Frontend Command = npm run dev url = http://localhost:5173 Backend command = npm run dev url =	All the server should run without any issue	All the three servers run successfully without any issue.	Pass

		http://localhost:8848 Prediction Model Command = python server.py url = http://localhost:5000			
--	--	--	--	--	--

Table 4.2 Test Case for User Registration

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC2.1	Navigating to home and displaying home page.	Enter frontend url http://localhost:5173	Home page should be displayed with navigation bar.	Home page is shown with navigation bar to navigate to different page.	Pass
TC2.2	Navigating to register page	Click on register tab on navbar	Registration page with register form should be displayed	Registration page with register form was be displayed	Pass
TC2.3	Register with filling required fields.	Name: Sanjay Khadgi Email: sanjaykhadgi9861@gmail.com Password: User123@ Contact: 9861365442	Redirecting to verification page	User verification page was displayed.	Pass

TC2.4	Register with incorrect or empty fields	Name: Email: 2212 Password: tc1 Contact: abd	Error Message Must be displayed.	Error: Please enter required fields. Error: Invalid details	Pass
-------	---	---	-------------------------------------	--	------

Table 4.3 Test Case for User Verification

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC3.1	Navigating to verification page	Email: sanjaykhadgi9861@gmail.com	After successful registration and OTP should be sent to user's email and user should be navigated to verification page	An OTP was received in user email and was redirected to verification page.	Pass
TC3.2	Verify with correct details.	Email: sanjaykhadgi9861@gmail.com OTP: 612374	An alert with success should be shown and user must be redirected to login page.	Verification successful. User was redirected to login page.	Pass
TC3.3	Verify with empty details.	Email: OTP:	Error Message Must be displayed.	Error: Please enter required fields.	Pass

TC3.4	Verify with incorrect email and password	Email: abc@abc.com OTP: tacs1	Error Message Must be displayed.	Error: Invalid OTP or Email	Pass
TC3.5	Verify with previously used OTP	Email: sanjaykhadgi9861@gmail.com OTP: 612374	Error Message must be displayed.	Error: Invalid OTP or Email	Pass

Table 4.4 Test Case for User Login

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC4.1	Navigating to login page	Click on login tab on navbar	Login page with login form should be displayed	Login page with login form was displayed.	Pass
TC4.2	Login with correct email and password	Email: sanjaykhadgi9861@gmail.com Password: User123@	Should be redirected to respective dashboard.	Redirected to user dashboard.	Pass
TC4.3	Login with empty fields	Email: Password:	Error Message Must be displayed.	Error: Please enter required fields.	Pass
TC4.4	Login with incorrect email and password	Email: abc@abc.com Password: tc1	Error Message Must be displayed.	Error: Invalid Credentials	Pass

Table 4.5 Test Case for Browsing House Lists

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC5.1	Navigating to House listing page	Click on house listings tab on navbar	House listings page with all houses should be displayed for both authenticated and unauthenticated users	Both unauthenticated and authenticated users can see house listings in house listings page.	Pass
TC5.2	Search house details by entering location name,	Location: Tinkune, Kathmandu Name: House in sales	List of houses of that location or name should be displayed.	Houses with input location or name. was displayed	Pass
TC5.3	Filter House by Location, Bedrooms, Floors, kitchen, etc.,	Bedrooms: 4 Floors: 3 Bedrooms: 4 Bathrooms: 2 Kitchen: 2 Location: Hebbal	List of houses with that filters should be displayed.	Houses with selected features was displayed.	Pass
TC5.4	View house details	Select a house to see more details	Selected house details must be displayed.	House details with more information was displayed.	Pass

Table 4. 6 Test Case for Predicting House Price

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC6.1	Navigating to House Price Prediction page authenticated user	Click on house Price Prediction tab on navbar only available for authenticated user.	House price prediction page should be displayed with input parameter form	House price prediction page was displayed with a prediction form for authenticated user.	Pass
TC6.2	Navigating to House price prediction page unauthenticated user	Try to access the prediction page without authenticating	House price prediction tab should not be displayed for unauthenticated user.	House price prediction tab was not shown for unauthenticated users	Pass
TC6.3	Predict Price by entering details	Area (sq.ft): 3369 Location: Hebbal Bath: 2 BHK: 2	Predicted price should be displayed and house with similar features should be recommended to users.	Predicted price was displayed and houses with similar features was recommended to authenticated users.	Pass
TC6.4	Predict Price by entering empty details	Area (sq.ft): Location: Bath: BHK:	Error message must be displayed.	Error: Please enter all the required fields	Pass

TC6.5	Predict Price by entering incorrect details	Area (sq.ft): abc Location: 123 Bath: asd BHK: dsa	Error message must be displayed.	Error: Please enter correct details	Pass
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Table 4. 7 Test Case for Admin Login

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC7.1	Navigating to login page	Click on login tab on navbar	Login page with login form should be displayed	Login page with login form was be displayed.	Pass
TC7.2	Login with correct email and password	Email: admin@gmail.com Password: Admin123@	Should be redirected to respective dashboard.	Redirected to admin dashboard.	Pass
TC7.3	Login with empty fields	Email: Password:	Error Message Must be displayed.	Error: Please enter required fields.	Pass
TC7.4	Login with incorrect email and password	Email: abc@abc.com Password: tc1	Error Message Must be displayed.	Error: Invalid Credentials	Pass

Table 4. 8 Test Case for Registering House Details

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC8.1	Navigating to House adding page	After successful admin authentication.	Add house page with a	Add house page with a	Pass

		Click on Add house link from admin dashboard.	form should be displayed	form was be displayed.	
TC8.2	Add house details by entering all required fields	Price: 30000000 Location: Tinkune, Kathmandu, Area Sq.ft : 3360 BHK: 3 Floors: 2.5 Amenities: parking	After successful submission alert should be displayed	An alert with saying House added successfully was displayed.	Pass
TC8.3	Add house details with empty fields	Price: Location: Area Sq.ft.: BHK: Floors: Amenities:	Error Message Must be displayed.	Error: Please enter required fields.	Pass
TC8.4	Add house details with incorrect details	Price: apple Location: 123 Area Sq.ft.: abc BHK: abc Floors: abc Amenities: 1	Error Message Must be displayed.	Error: Please enter fields correctly.	Pass

Table 4. 9 Test Case for Training the Model

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC9.1	Navigating to train model page	Click on Train model	Train model page with an	Train model page with	Pass

		link in admin dashboard	input field to take .csv file should be displayed.	input field to take .csv file was be displayed.	
TC9.2	Train the model with pre-processed datasets and check accuracy.	File Houseprices.csv	Model should be trained and accuracy rate should be displayed on the screen	Model was trained and accuracy rate was displayed on the screen.	Pass
TC9.4	Dump the trained model in the pickle file to predict price.	Model_train.d.pkl	.pkl file should be created and must be used to predict the price and old file should be deleted.	.pkl file was created and was used to predict the new prices for the house.	Pass
TC9.5	Train the model with unprocessed datasets or different datasets which has more feature than the model requires	File: realestate.csv	Model should be trained and accuracy rate should be displayed on the screen.	Model was not trained and an error occurred.	Fail

4.2.2 Test Case for System Testing

Table 4.10 Black Box System Testing

S.N.	Description	Requirements	Result	Test Status
1.	Authentication	User provides valid registration details	Users can create new account and sign in to the	Passed

		(name, email, password, etc.,	system to predict the house prices.	
2.	View Houses	Search query, filter parameters, (location, floors, bedrooms, etc.,)	Both unauthenticated and authenticated users can view the house list and house details	Passed
3.	Predict House Price	Authentication with correct details and prediction parameters (location, area sq.ft., bhk etc.,)	Authenticated users can predict house prices based on the parameters they provide and get price prediction as well as get recommended house based on that features.	Passed
4.	Add Houses	Admin role with admin login credentials and house details. (eg., bhk, area, floors, location etc.,)	Admin can add, update, view and delete house details.	Passed
5.	Manage Users	Admin role with admin login credentials and user details. (roles)	Admin can update user roles from admin to user and vice versa.	Passed
6.	Pre-process datasets	Admin role with admin login credentials and datasets uploaded.	Admin can remove outliers, extract features, reduce dimensionality from datasets.	Passed
7	Train Model	Admin role with admin login credentials, datasets and learning models.	Admin can train the model by feeding pre-processed dataset and check the accuracy score.	Passed

4.3 Result Analysis

4.3.1 Evaluating Accuracy

Accuracy is a commonly used metric in machine learning to evaluate the model's performance. Accuracy is the proportion of correctly classified instances in the datasets. To calculate accuracy, divide the datasets into two parts: training and testing sets. The training set trains the model, while the testing set evaluates its performance. For Linear Regression, Random Forest and KNN models the most commonly used accuracy measures are:

Precision: Precision refers to the number of true positive instances out of the predicted ones. In the context of real estate house price prediction, precision is referred as the ratio of accurately predicted price categories (true positives) to all instances predicted as positive. A high precision value of 83% is obtained from the model, which indicates that the model's prediction for a specific price category is like to be true.

Recall (Sensitivity): Recall determines how many actual positive cases were correctly predicted by the model. In real estate house price prediction, recall is the ratio of correctly predicted price categories (true positives) to all real positive's cases (true positives + false negatives). With a recall of 83% is obtained, which denotes the model effectively captured a significant percentage of true positive cases in the dataset.

F1-Score: The F1-Score is a balanced metric that combines precision and recall, performing as the harmonic mean of the two measures. The F1-Score of 83% represents the trade-off between precision and recall. This is especially useful when evaluating model's performance in the presence of both false positives and false negatives.

Accuracy: Accuracy is calculated as the ratio of accurately predicted instances to the total number of instances. The model obtained an accuracy of 83% which means that around 83% of predictions were right. However, accuracy alone may not be the most reliable metric in the case of imbalanced datasets.

4.3.2 Evaluating Accuracy of the Models

Table 4. 11 House Price Prediction Classification Report Using Linear Regression

House Price Prediction Report				
	Precision	Recall	F1-Score	Support
	82%	82%	82%	1469
Accuracy			82%	1469

Macro Average	77%	74%	75%	1469
Weighted Average	82%	82%	82%	1469

The Linear Regression model performed well, achieving an accuracy of 83%, with high precision, recall and F1-score, all also at 83%. These metrics indicate that this model can constantly predict house prices that are close to the actual value across various price ranges. The macro average scores (77% precision, 74% recall, 75% F1-score) show that while there is some variation in performance across different price categories, it is generally consistent. The weighted average, which considers all predictions together, also confirms this model's strong performance with an 83% score. This concludes that Linear Regression is effective for predicting house price and is the best-performing model in comparison.

Table 4. 12 House Price Prediction Classification Report Using Random Forest Regressor

House Price Prediction Report				
	Precision	Recall	F1-Score	Support
	82%	82%	82%	1469
Accuracy			82%	1470
Macro Average	75%	74%	74%	1470
Weighted Average	83%	82%	82%	1470

The Random Forest Regressor however, did not perform well. It achieved an accuracy of only 25%, with precision at 36% and a low F1-score of 10%. These values indicate that this model has some difficulty predicting house prices correctly within the defined categories. The low F1-score, in particular, suggest that the model may be underfitting the dataset, leading to inconsistent price predictions. Also, the macro and weighted averages, which are also low (18% precision, 25% recall, 10% F1-score), show that the Random Forest regressor model performance is unreliable across all price ranges. Overall, this model does not provide accurate predictions for house prices.

Table 4. 13 House Price Prediction Classification Report Using KNN Regressor

House Price Prediction Report				
	Precision	Recall	F1-Score	Support

	77%	77%	77%	1469
Accuracy			77%	1469
Macro Average	69%	67%	67%	1469
Weighted Average	77%	77%	77%	1469

The KNN Regressor performed similar to the Random Forest Regressor, with an accuracy of only 25%. Its precision was especially low at 6% meaning it struggled significantly to make accurate predictions. Recall was 25% and the F1-score was low at 10%, indicating that the KNN model could not predict prices effectively. This model's poor results indicate that the model may be underfitting the dataset or it cannot track the continuous nature of house prices. The low macro and weighted average scores confirm that KNN regressor does not work well for this task, showing consistent inaccuracy across all categories.

K-Fold Cross validation technique

The tables below summarize the results of a K-Fold cross validation process for three different regression models: Multiple Linear Regression, Random Forest Regressor, and K-Nearest Neighbours (K-NN) Regressor. K-Fold cross validation is a technique used to evaluate the performance models by splitting the data into multiple subsets of "folds". For each fold, a portion of the data is used to train the model, and the remaining portion is used to test it. This helps to give a balanced view of how well the model performs across different portions of the data.

In the tables below, the data is split into five different folds, with various training and testing sizes. Each fold has different percentage of data used for training and testing, which affects the model's accuracy and error metrics.

Table 4. 14 Cross Validation Results for K-Fold Size=1

Train Size = 70%	5143			
Test Size = 30%	2205			
Regression Model Algorithms	MSE	RMSE	MAE	R-squared
Multiple Linear Regression	1605.87	40.07	18.06	0.82
Random Forest Regressor	2352.59	48.50	19.65	0.73
K-NN Regressor	2786.52	52.79	24.41	0.68

In this first fold with 70% of the data used for training, Multiple Linear Regression performs best, achieving the lowest Mean Squared Error (MSE) of 1605.87 and an R-squared value of 0.82, meaning it can explain 82% of the variance in the data. Random Forest and K-NN show higher errors of 2352.59 and 2786.52, and has the R-squared of 0.73 and K-NN being the lowest of 0.68, indicating low accuracy

Table 4. 15 Cross Validation Results for K-Fold Size=2

Train Size = 50%	3674			
Test Size = 50%	3674			
Regression Model Algorithms	MSE	RMSE	MAE	R-squared
Multiple Linear Regression	1472.42	38.37	18.54	0.84
Random Forest Regressor	2024.30	44.99	20.20	0.77
K-NN Regressor	2804.61	52.96	25.62	0.69

In the second fold, where training and testing data are evenly split, Multiple Linear Regression again achieves the best results with an MSE of 1472.42 and an R-squared of 0.84. Random Forest performs moderately well with an R-squared of 0.77, while K-NN shows higher error rates and a lower R-squared of 0.69.

Table 4.16 Cross Validation Results for K-Fold Size=3

Train Size = 67%	4923			
Test Size = 33%	2425			
Regression Model Algorithms	MSE	RMSE	MAE	R-squared
Multiple Linear Regression	1563.87	39.55	18.22	0.82
Random Forest Regressor	2305.68	48.02	19.73	0.73
K-NN Regressor	2672.90	51.70	24.36	0.69

In the third fold, with training split 67%, Multiple Linear Regression again has high score with an MSE of 1563.87 and R-squared of 0.82. Random Forest and K-NN got lower accuracy with higher errors, with K-NN having an R-squared of only 0.69 and Random Forest with an R-squared of 0.73.

Table 4.17 Cross Validation Results for K-Fold Size=4

Train Size = 75%	5511			
Test Size = 25%	1837			
Regression Model Algorithms	MSE	RMSE	MAE	R-squared

Multiple Linear Regression	1707.86	41.33	17.98	0.81
Random Forest Regressor	2718.29	52.14	19.35	0.70
K-NN Regressor	3020.88	54.96	23.99	0.67

In the fourth fold, with 75% of data for training, Multiple Linear Regression remains the best performer, but with slightly higher MSE of 1707.86 and a small drop in R-squared to 0.81. Random forest and K-NN still got higher errors of 2718.29 and 3020.88 MSE and R-square of 0.70 and 0.67.

Table 4.18 Cross Validation Results for K-Fold Size=5

Train Size = 80%	5878			
Test Size = 20%	1470			
Regression Model Algorithms	MSE	RMSE	MAE	R-squared
Multiple Linear Regression	1008.42	31.76	17.69	0.84
Random Forest Regressor	1342.31	36.64	18.61	0.78
K-NN Regressor	1793.21	42.35	23.24	0.71

In the fifth fold, with the highest training data percentage of 80%, Multiple Linear Regression achieves its lowest MSE of 1008.42 and the highest R-squared of 0.84, showing improved accuracy. Random forest performs moderately with an MSE of 1342.31 and R-square of 0.78, while K-NN again has the highest error metrics of 1793.21 and a lower R-squared score of 0.71.

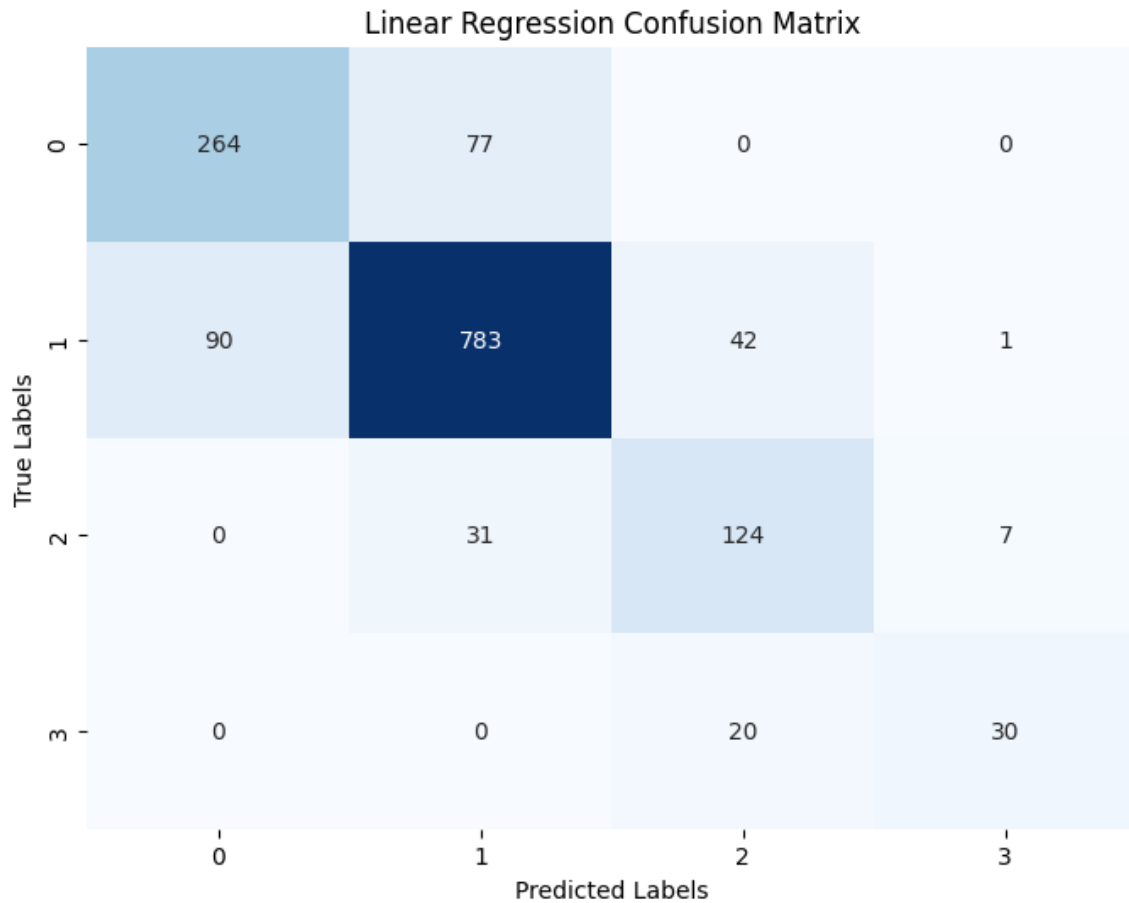


Figure 4. 8 Confusion matrix for Multiple Linear Regression

The confusion matrix above visualizes the performance of the Multiple Linear Regression model in predicting house prices across four price ranges the diagonal values indicate the correct prediction, with 264 instances in cheap price range (category 0), 783 instances in the moderate price range (category 1), 124 instances in expensive price range (category 2), and 30 instances in luxury price range (category 3) predicted accurately. This demonstrates that the model is particularly effective in predicting the moderate and cheap price ranges, as seen from the figure above with high counts of correct prediction in these categories. However, there are some mispredictions. For example, 77 instances of actual prices in the cheap range are predicted in the moderate range, while 90 instances in the moderate range are predicted as cheap price range. 42 instances in the moderate range are mis predicted as the expensive range, and 1 instance is mis predicted in the luxury range. In the expensive range, 31 instances are incorrectly predicted as moderate, and 7 instances are predicted as belonging to the luxury price range. The luxury price range has 20 instances predicted as the expensive range, indicating some difficulty for the model in distinguishing this category.

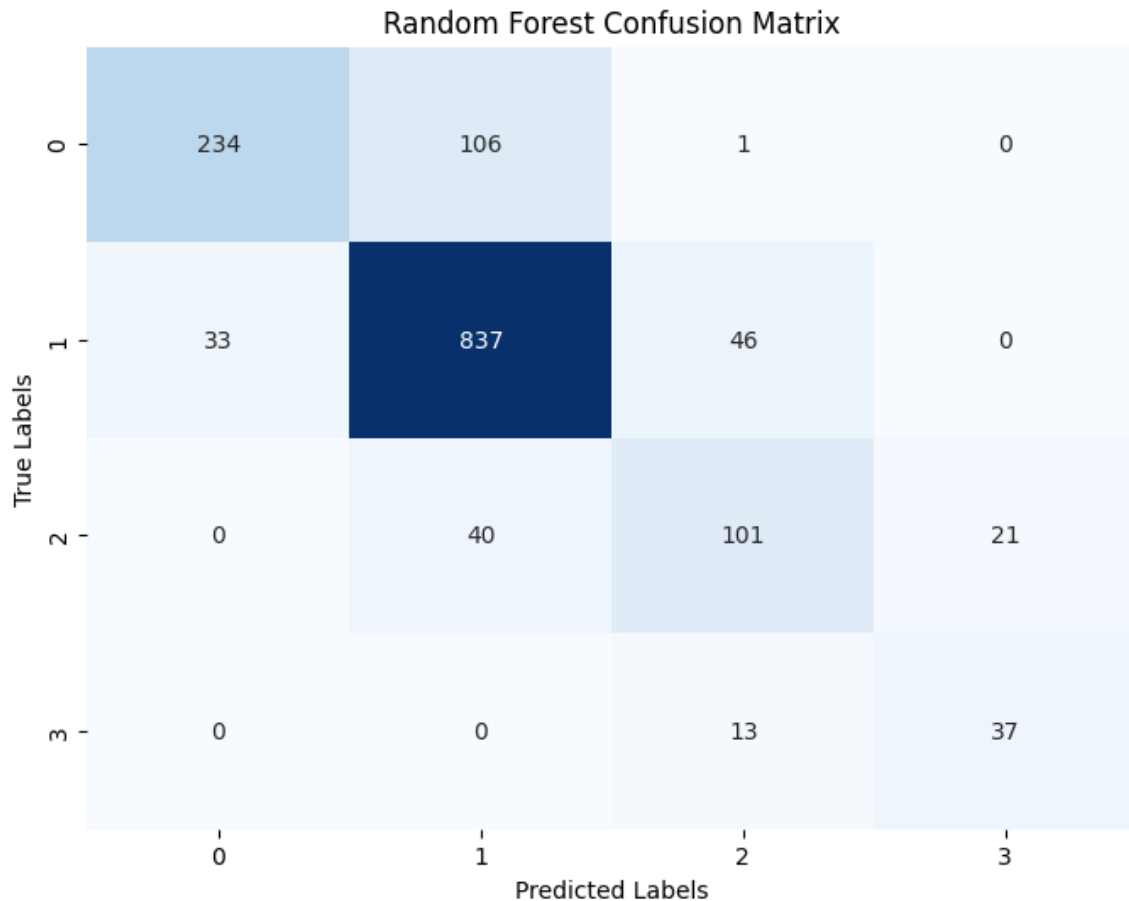


Figure 4. 9 Confusion matrix for Random Forest Regressor

The confusion matrix above illustrates the performance of the Random Forest Regressor model in predicting house price range across four price categories. The diagonal values represent accurate prediction, with 245 instances in the cheap price range, 837 instances in the moderate range, 101 instances in the expensive range, and 37 instances in the luxury range predicted correctly. This indicates that the model is effective at distinguishing between the moderate and cheap price ranges, as shown by the higher number of accurate predictions in these categories.

Some mispredictions are also present in the model. For example, 106 instances of actual price in the cheap range are mis predicted in the moderate range, and 33 instance in the moderate range are predicted as the cheap range. 46 instances in the moderate range are mis predicted in the expensive range, while 40 instances in the expensive range are predicted in the moderate range, 21 instances are predicted in the luxury range. While the luxury price range has more correct prediction than mis predictions, 13 instances in this range are predicted as the expensive range.

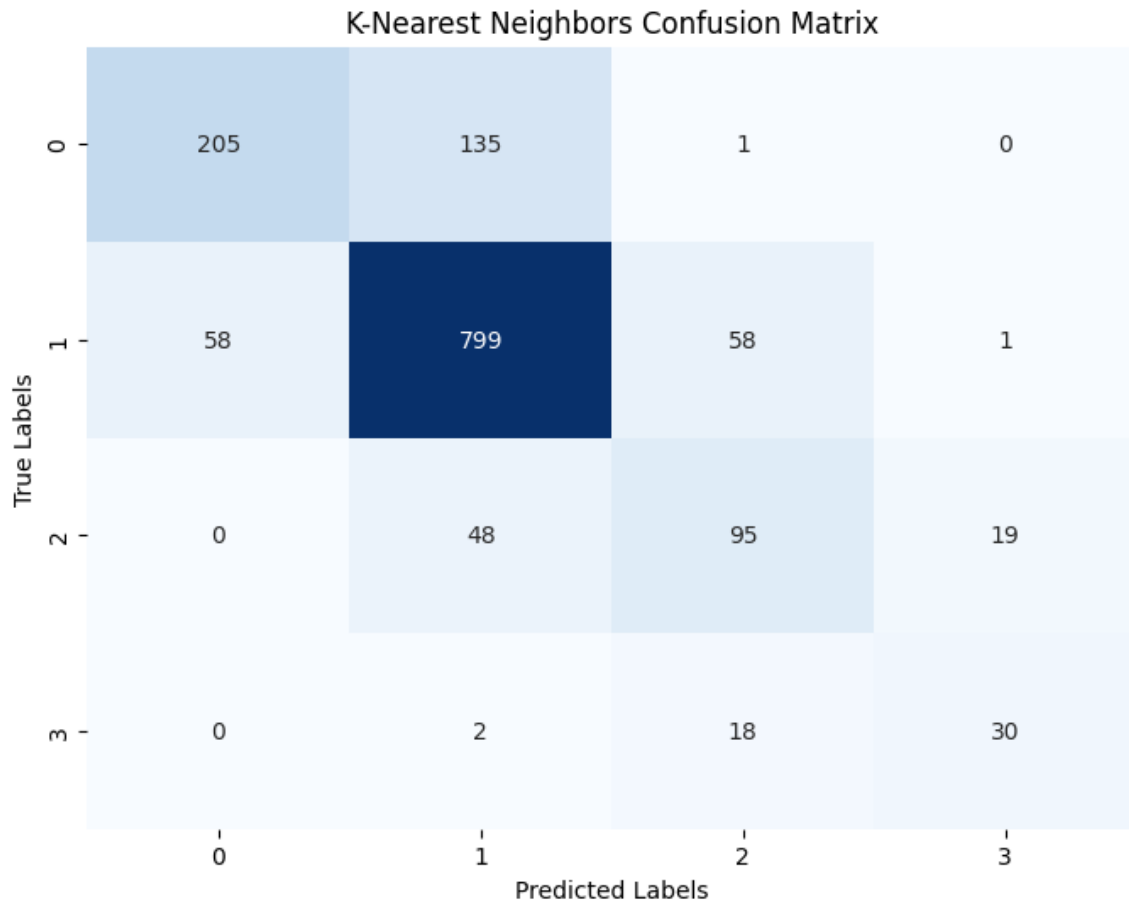


Figure 4. 10 Confusion matrix for KNN Regressor

The confusion matrix above shows the performance of the K-Nearest Neighbour (KNN) model in predicting house prices across four prices ranges. The diagonal values are correct predictions, with 205 instances in the cheap price range, 799 instances in the moderate price range, 95 instances in the expensive price range, and 30 instances in the luxury price range predicted accurately. KNN model performs well in predicting the moderate price range, as it has the highest number of correct predictions in this category.

Although, there are some mis predictions in this model. For example, 135 instances in the cheap price range are mis predicted as moderate range, and 58 instances in the moderate are mis predicted as both cheap and expensive ranges. In the expensive price range, 48 instances are incorrectly predicted as moderate, and 19 instances are predicted as luxury range. The luxury price range shows 18 instances incorrectly predicted as the expensive range, while only 2 instances are mis predicted as the moderate range.

CHAPTER 5

CONCLUSION AND FUTURE RECOMMENDATION

5.1 Conclusion

The project “Real Estate House Price Prediction” is a project aiming to develop a model that can predict the price of the house based on various features like area (sq. ft.), location, BHK, bathrooms, etc., Throughout this project, datasets were collected from different sites like Kaggle.com. For accurate data various steps like data cleaning, feature extraction and different model was built. Multiple Linear Regression model was written from scratch and other model like Random Forest and KNN Regressor were imported from the scikit-learn library to compare the accuracy between models. The model was trained using different datasets and tested if it can predict the price accurately.

This system can be viewed by any user. Unauthenticated user can only view house lists and house details, search and filter different house, while authenticated user can predict the house price and get recommendation of house with similar features based on their input parameters. These prediction and recommendations help user make a better decision based on current market trends and property analysis.

5.2 Future Recommendations

The “Real estate price prediction” which is currently in experimental stage, has the potential to become a powerful tool for predicting house price in the real estate business. A number of changes can be made to increase its accuracy and usability. One key improvement would be expanding the feature set to include additional features like amenities, marketplaces, schools and neighbourhood quality. These features alongside dataset with complex features could be processed without needing much pre-processing would help in feeding model to learn from different categories of datasets.

In the future, the system could be enhanced by allowing users to add their own properties for sale, creating more interactive system where buyers and sellers can engage directly. Furthermore, a mobile application could enhance accessibility, giving the users to browser listing, predict prices and receive house recommendations from anywhere. By adding these features, the system could not only server as a price prediction tool but also evolve into a comprehensive real estate platform that supports users in both buying and selling properties.

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House Listing - User Role

Vite - React

localhost:5173/properties

Real Estate House Price Prediction

HomeHouse ListingsPredict Price

Search Property

Locations:

Bhaktapur, Sallaghari

Bhaktapur, Suryabinayak

Kathmandu, Kalanki

Kathmandu, Tinkune

Lalitpur, Imadol

Lalitpur, Lubhu

Floors:

0

1

2

3

4

5

Bedrooms:

0

1

2

3

4

5

Kitchen:

0

1

2

3

4

5

Rs. 19400000

Gapali, Bhaktapur - घर दि...

Bhaktapur, Suryabinayak

5 Bed2 Bath1 Kitchen

VIEW DETAILS

Rs. 5775000

Mero City Apartment : Apa...

Bhaktapur, Sallaghari

3 Bed2 Bath1 Kitchen

VIEW DETAILS

Rs. 30000000

Property on sale....

Lalitpur, Imadol

4 Bed4 Bath2 Kitchen

VIEW DETAILS

Rs. 20000000

Residential House On Sale....

Lalitpur, Lubhu

3 Bed2 Bath1 Kitchen

VIEW DETAILS

Rs. 30000000

Rose Villa Phase II....

Bhaktapur, Suryabinayak

3 Bed3 Bath3 Kitchen

VIEW DETAILS

Rs. 31200000

Updated Property....

Lalitpur, Lubhu

4 Bed1 Bath2 Kitchen

VIEW DETAILS

Predict Price - User Role

← → ↻

localhost:5173/predict-price

☆

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Real Estate House Price Prediction

Home House Listings Predict Price 

Predict Price

Area (sq. ft.) *

2850

Select BHK:

☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Select Bathrooms:

☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Select Locations

1st Block Jayanagar

Predict Price

Predicted Price *

Rs. 23400000



Rs.	Rs.	Rs.
35000	30000	31200
Best Proper For Sale...	Proper on sale....	Update Proper
1st	Laltpu	Lubhu
Imadol		4

View Profile – Both Role

Admin

admin@gmail.com

+977 9861365442

Kathmandu, Baneshwor

Name *

Admin

Email Address *

admin@gmail.com

Phone No.

+977 9861365442

Address

Kathmandu, Baneshwor

Browse...

No file selected.

SUBMIT

Update Password

Old Password *

New Password *

SUBMIT

View Users – Admin Role

Morgan

REAL ESTATE

Property

Locations

Users

Train Model

Logout

All Users

See Information of (3 users)

S.N.	User Details	Address	Verified	Role	Joined Date	Actions
1	test user test@gmail.com +977 9861365442	Kathmandu, Tinkune	No	USER	2024-10-19T12:20:51.034Z	
2	Sanjay Khadgi sanjay@gmail.com +977 9861365442	Lubhu, Lalitpur	Yes	USER	2024-10-16T11:40:36.372Z	
3	Admin admin@gmail.com +977 9861365442	Kathmandu, Baneshwor	Yes	ADMIN	2024-10-16T07:54:12.550Z	

Train Model – Admin Role

Morgan

REAL ESTATE

Property

Locations

Users

Train Model

Logout

Train Model

Fill required (*) fields

Upload Datasets*

Browse...

bengaluru_house_prices.csv

SUBMIT

Add Property – Admin Role



Property

Locations

Users

Train Model

Logout

Add Property

Fill required (*) fields

Property Title *

Best Property For Sale

Price *

30000000

Area (sq.ft.) *

2850

Select Location

Bhaktapur, Sallaghari

Bedrooms:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Kitchen:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Bathrooms:

☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

All Properties – Admin Role



Property

Locations

Users

Train Model

Logout

All Properties

See information of (6 - properties)

S.N.	Property Details	Location	Area (sq.ft.)	Price	Floors	Bedrooms	Bathrooms	Kitchen
1	 Gapali, Bhaktapur - gapali-bhaktapur-hou	Bhaktapur, Suryabinayak	1024	19400000	2	5	3	1
2	 Mero City Apartment mero-city-apartment-	Bhaktapur, Sallaghari	1122	5775000	5	3	2	1
3	 Property on sale property-on-sale	Lalitpur, Imadol	4000	30000000	3	4	4	2
4	 Residential House On residential-house-on	Lalitpur, Lubhu	3500	20000000	2	3	2	1
5	 Rose Villa Phase II rose-villa-phase-ii	Bhaktapur, Suryabinayak	3200	30000000	3	3	3	3
6	 Updated Property updated-property	Lalitpur, Lubhu	3024	31200000	1	4	1	2